

Integrating Genomic Data: LD and SNP-based annotation

HUGEN 2073

Genomic Data Visualization and Integration

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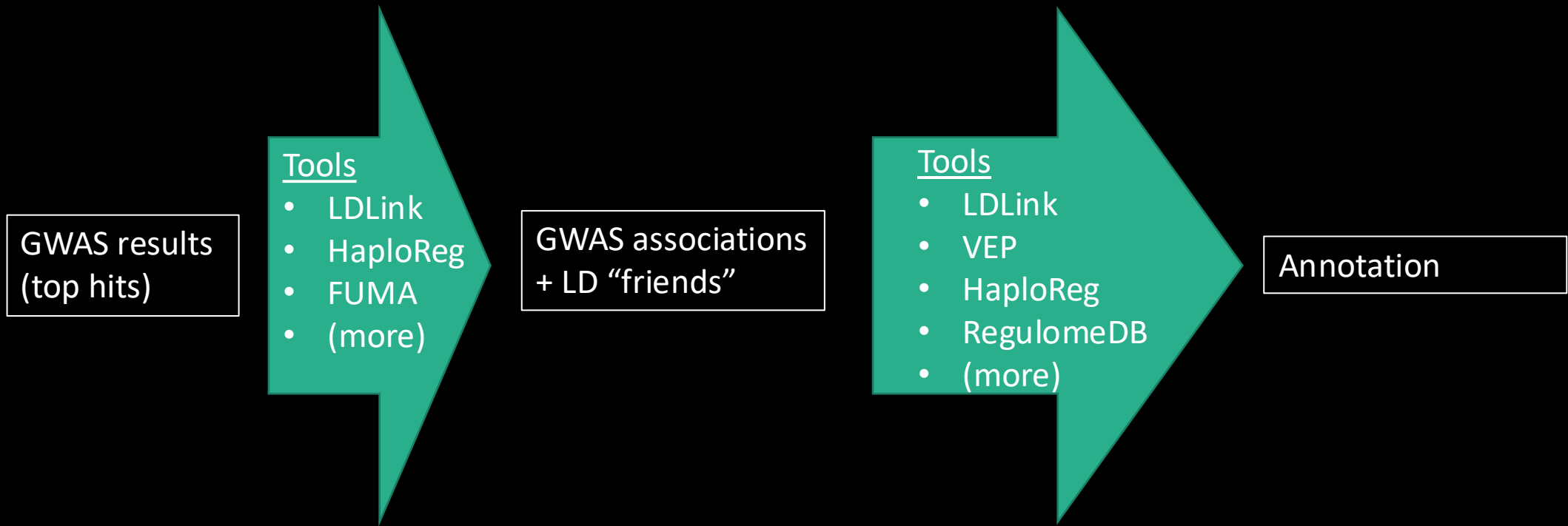
Last time

- Why annotate?
- Linkage disequilibrium
- Overview of some tools/databases
- Introductory examples
 - FUMA output
 - Successes, e.g., in the *IRX3/IRX5* BMI “story”
- Limitations/caveats

This lecture

- More detailed walkthrough for querying LD information and SNP-level annotation
 - LDlink
 - VEP
 - HaploReg
 - RegulomeDB
 - GWASCatalog
 - Using R packages to access these data

Possible “workflow”



- Note that these tools have some overlapping functions and some unique functions
- The output from one tool may be the input into another (or the input for a different function of the same tool)

LDlink

<https://ldlink.nci.nih.gov/?tab=home>

LDlink is suite of web-based tools for looking up and visualizing LD information, using dbSNP, 1000Genomes, GTEx data (and more)

There is an R package API, LDlinkR, which you'll try out.

LDassoc

Interactively visualize association p-value results and linkage disequilibrium patterns for a genomic region of interest.

LDexpress

Search if a list of variants (or variants in LD with those variants) is associated with gene expression in multiple tissue types.

LDhap

Calculate population specific haplotype frequencies of all haplotypes observed for a list of query variants.

LDmatrix

Create an interactive heatmap matrix of pairwise linkage disequilibrium statistics.

LDpair

Investigate correlated alleles for a pair of variants in high LD.

LDpop

Investigate allele frequencies and linkage disequilibrium patterns across 1000G populations.

LDproxy

Interactively explore proxy and putatively functional variants for a query variant.

LDtrait

Search if a list of variants (or variants in LD with those variants) have previously been associated with a trait or disease.

SNPchip

Find commercial genotyping platforms for variants.

SNPclip

Prune a list of variants by linkage disequilibrium.

LDProxy

- Input: SNP (and populations)
- Output: Downloadable list of SNPs in LD with annotation and plot

Proxy Variants

Show entries

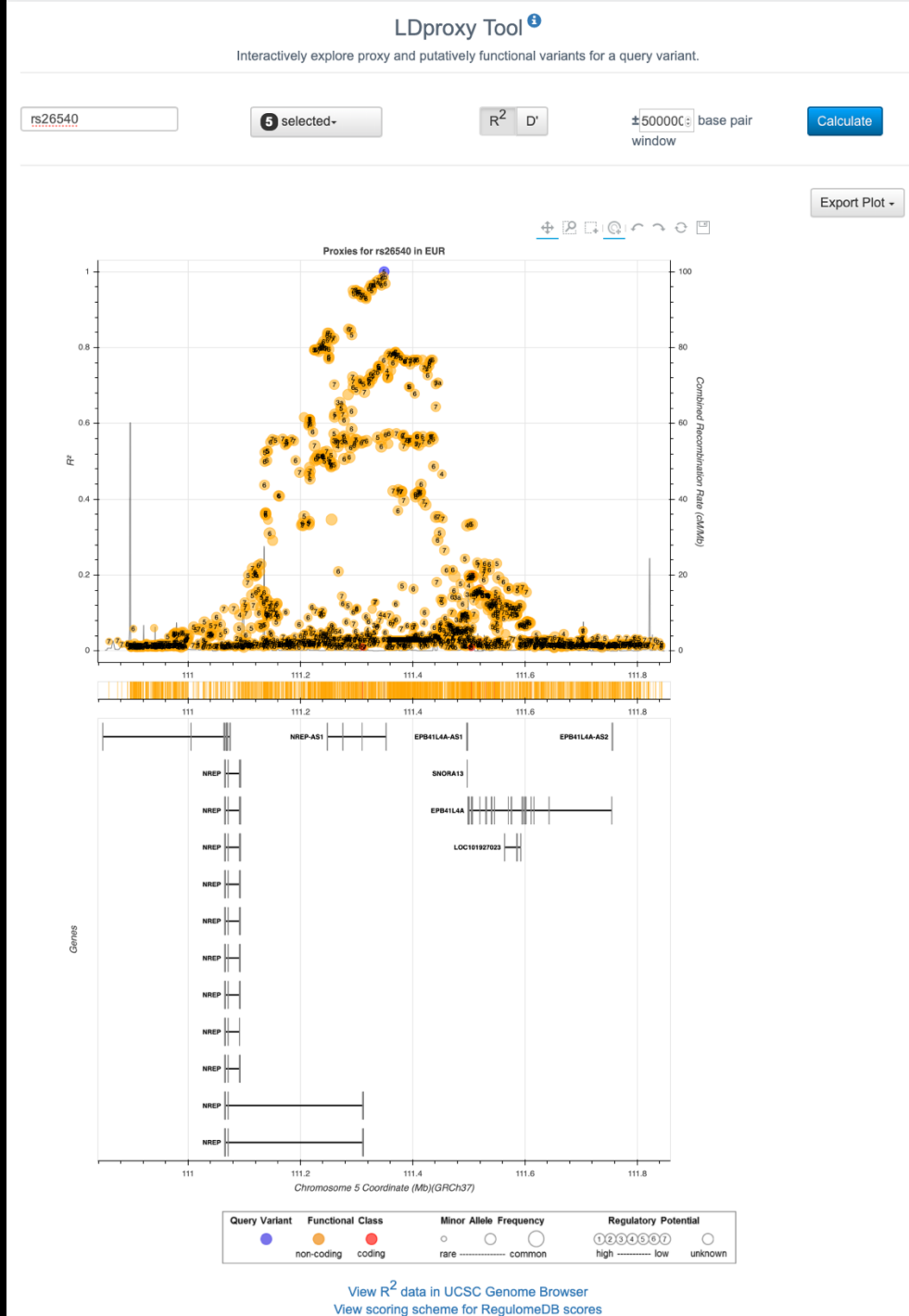
Search:

RS Number	Chr	Position (GRCh37)	Alleles	MAF	Distance	D'	R ²	Correlated Alleles	RegulomeDB	HaploReg	Functional Class
rs26542	5	111348069	(T/G)	0.3191	1251	1.0	0.9864	T=T,C=G	2b		NA
rs255930	5	111345538	(C/T)	0.3201	3782	0.9954	0.9819	T=C,C=T	6		NA
rs26560	5	111338466	(G/C)	0.3211	10854	0.9909	0.9774	T=G,C=C	7		NA
rs34942	5	111334801	(G/A)	0.3201	14519	0.9908	0.9728	T=G,C=A	3a		NA
rs26539	5	111351359	(G/A)	0.3151	2039	1.0	0.9685	T=G,C=A	6		NA
rs255928	5	111342333	(A/G)	0.3171	6987	0.9954	0.9684	T=A,C=G	6		NA
rs34936	5	111325083	(T/C)	0.3241	24237	0.9863	0.964	T=T,C=C	6		NA
rs26556	5	111328992	(A/G)	0.3241	20328	0.9863	0.964	T=A,C=G	3a		NA
rs26559	5	111329649	(C/T)	0.3241	19671	0.9863	0.964	T=C,C=T	7		NA
rs34941	5	111330141	(A/C)	0.3241	19179	0.9863	0.964	T=A,C=C	5		NA

Showing 1 to 10 of 2,191 entries

[Download all proxy variants](#)

Previous 2 3 4 5 ... 220 Next



LDtrait

- Input: SNPs of interest
- Output: GWAS associations for the input SNPs *and* their LD "friends"

LDtrait Tool

Search if a list of variants (or variants in LD with those variants) have previously been associated with a trait or disease.

rs13107325

Upload file with variants

Browse

7 selected

R²

D'

Thresholds

R²: 0.1

±50000 base pair window

Calculate

Variants in LD with GWAS Catalog

RS Number

rs13107325

Show 10 entries

Search:

Details for rs13107325

GWAS Trait	RS Number	Position (GRCh37)	Alleles	R ²	D'	LDpair	Risk Allele Frequency	Beta or OR	Effect Size (95% CI)	P-value	GWAS Catalog
Refractive error	rs13107325	chr4:103188709	C=0.998, T=0.002	1.0	1.0	link	NA	NA	NA	4x10 ⁻¹⁷	link
Risk-taking tendency (4-domain principal component model)	rs13107325	chr4:103188709	C=0.998, T=0.002	1.0	1.0	link	0.073	0.036	0.026-0.046	4x10 ⁻¹³	link
Low density lipoprotein cholesterol levels	rs13107325	chr4:103188709	C=0.998, T=0.002	1.0	1.0	link	NR	0.031	0.023-0.04	1x10 ⁻¹⁰	link
Triglyceride levels	rs13107325	chr4:103188709	C=0.998, T=0.002	1.0	1.0	link	NR	0.031	0.022-0.04	5x10 ⁻¹⁰	link
Hemoglobin levels	rs13107325	chr4:103188709	C=0.998, T=0.002	1.0	1.0	link	NA	0.048	NA	2x10 ⁻²⁴	link
Leisure sedentary behaviour (television watching)	rs13107325	chr4:103188709	C=0.998, T=0.002	1.0	1.0	link	NA	0.029	0.021-0.037	2x10 ⁻¹²	link
Apolipoprotein A1 levels	rs13107325	chr4:103188709	C=0.998, T=0.002	1.0	1.0	link	0.925	0.072	0.064-0.079	5x10 ⁻⁸²	link
Nucleus accumbens volume	rs13107325	chr4:103188709	C=0.998, T=0.002	1.0	1.0	link	0.064	6.150	NA	8x10 ⁻¹⁰	link
Brain region volumes	rs13107325	chr4:103188709	C=0.998, T=0.002	1.0	1.0	link	NR	NA	NA	2x10 ⁻¹⁸	link
Spherical equivalent	rs13107325	chr4:103188709	C=0.998, T=0.002	1.0	1.0	link	NA	0.130	0.091-0.169	3x10 ⁻¹⁰	link

Showing 1 to 10 of 247 entries

Previous

1

2

3

4

5

...

25

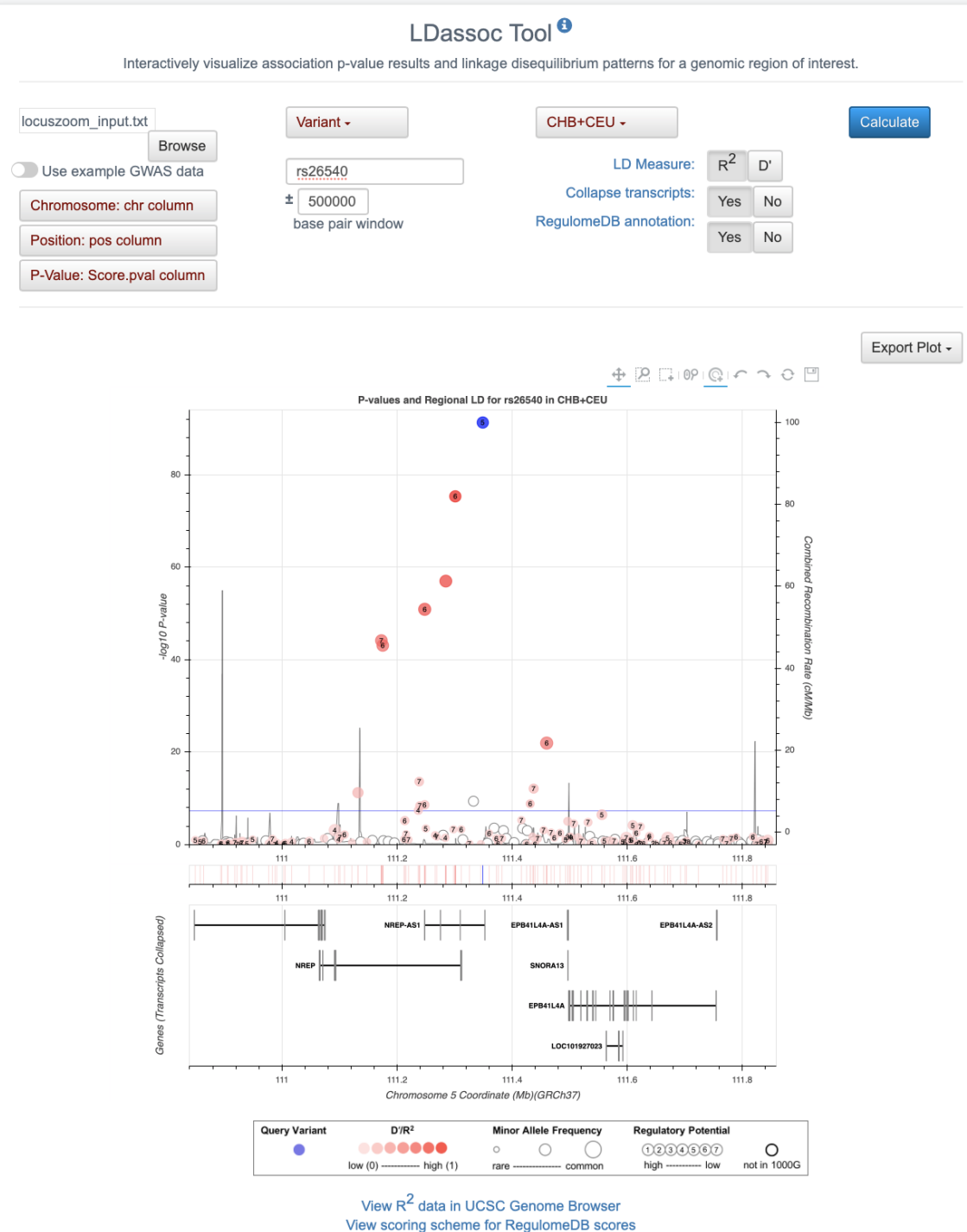
Next

Download GWAS Catalog annotated variant list

GWAS Catalog last updated on 3/15/2021, 01:04 AM (GMT-0400)

LDlink > LDassoc

- Input: GWAS results (select region/population)
- Output: Interactive LocusZoom plot with LD and allele frequency annotation



LDlink > LDassoc (con't.)

- Input: GWAS results (select region/population)
- Output: Interactive LocusZoom plot with LD and allele frequency annotation (with downloadable table)

Number of Individuals: 202
Variants in Region: 42
Run time: 1.71 seconds

Association Results

Show 10 entries

Search:

RS Number	Chr	Position (GRCh37)	Alleles	MAF	Distance	D'	R ²	Correlated Alleles	Association P-value	RegulomeDB	HaploReg	Functional Class
rs26540	5	111349320	(T/C)	0.3861	0	1.0	1.0	T=T,C=C	5.14188611821743e-92	5		NA
rs9326846	5	111301546	(A/G)	0.3837	47774	0.9685	0.9282	A=T,G=C	4.79109899374149e-76	6		NA
rs5870477	5	111285059	(T/-)	0.4356	64261	0.9659	0.7603	T=T,-=C	1.02100330402943e-57	.		NA
rs638424	5	111248399	(T/G)	0.4579	100921	0.9645	0.6927	T=C,G=T	1.30102008690746e-51	6		NA
rs9637887	5	111173095	(A/G)	0.4183	176225	0.7906	0.5467	A=T,G=C	7.39917808665142e-45	7		NA
rs6871530	5	111175425	(A/T)	0.4183	173895	0.7906	0.5467	A=C,T=T	8.44225698877688e-44	6		NA
rs77270851	5	111238805	(G/C)	0.151	110515	0.9575	0.1026	G=C,C=T	2.39247380156509e-14	7		NA
rs5870474	5	111132376	(TGAG/-)	0.2896	216944	0.7565	0.1468	TGAG=C,-=T	5.92698319465263e-12	.		NA
rs10515435	5	111248332	(A/G)	0.1436	100988	0.9553	0.0962	NA	2.59866143894834e-9	6		NA
rs35986445	5	111239756	(T/C)	0.1015	109564	1.0	0.071	NA	3.99863814721119e-9	7		NA

RS Results Chr Position (GRCh37) Alleles MAF Distance D' R² Correlated Alleles Association P-value RegulomeDB HaploReg Functional Class

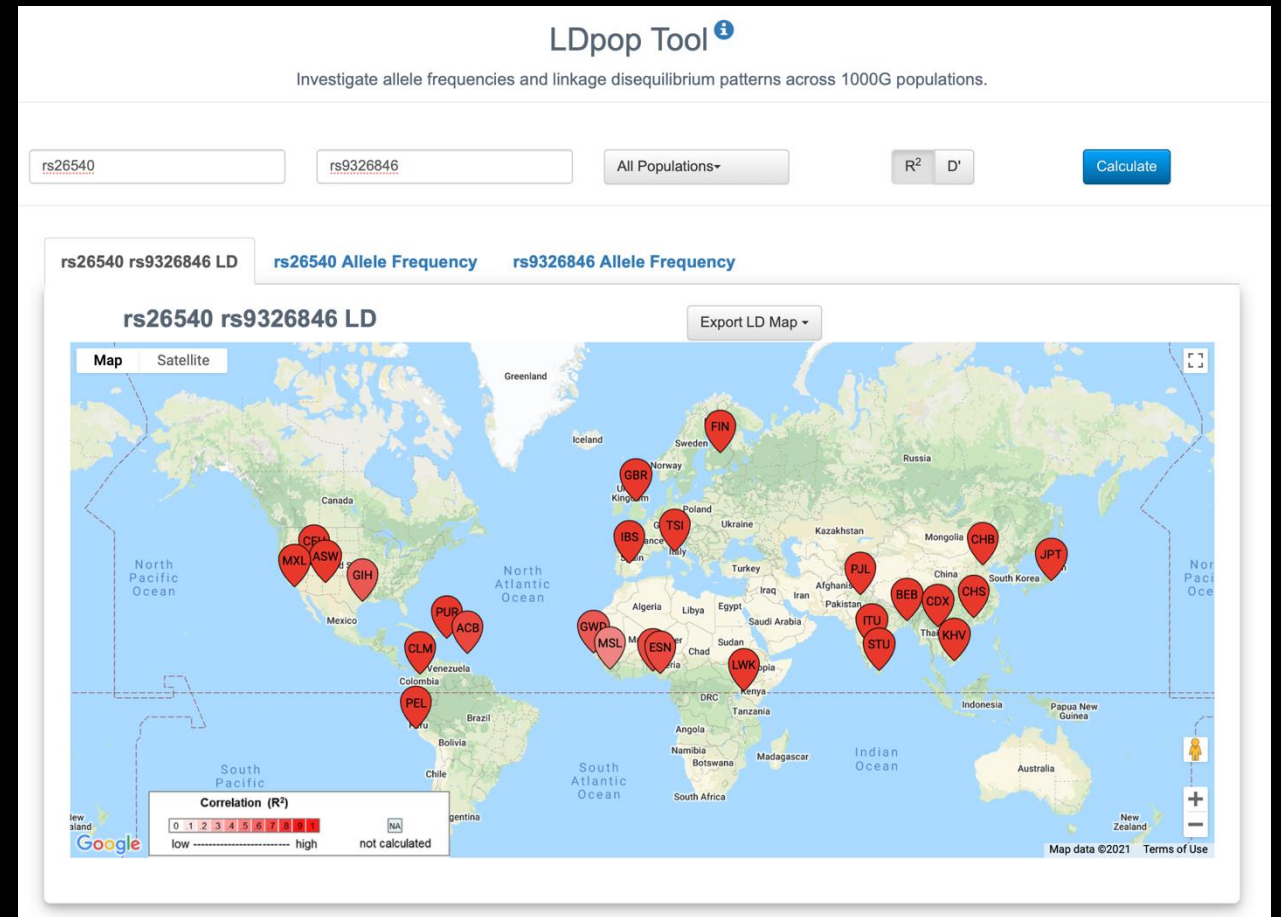
Showing 1 to 10 of 42 entries

Download association data for all variants

Previous 1 2 3 4 5 Next

LDlink > LDpop & LDPair

- Input: two SNPs
- Output: LD and allele frequency data for selected populations



LDlink > LDpop & LDPair

Showing 1 to 32 of 32 entries Search:

[Download Table](#)

Population	N	rs26540 Allele Freq	rs9326846 Allele Freq	R ²	D'	LDpair
ALL	2504	T: 53.39%, C: 46.61%	A: 54.15%, G: 45.85%	0.8773	0.9511	link
AFR	661	T: 62.18%, C: 37.82%	A: 61.42%, G: 38.58%	0.7603	0.886	link
YRI	108	T: 64.35%, C: 35.65%	A: 65.74%, G: 34.26%	0.8259	0.937	link
LWK	99	T: 54.55%, C: 45.45%	A: 55.56%, G: 44.44%	0.9204	0.9792	link
GWD	113	T: 67.7%, C: 32.3%	A: 64.16%, G: 35.84%	0.6815	0.8932	link
MSL	85	T: 58.24%, C: 41.76%	A: 51.18%, G: 48.82%	0.524	0.8349	link
ESN	99	T: 64.65%, C: 35.35%	A: 66.16%, G: 33.84%	0.7705	0.9076	link
ASW	61	T: 63.93%, C: 36.07%	A: 67.21%, G: 32.79%	0.8647	1	link
ACB	96	T: 60.94%, C: 39.06%	A: 59.9%, G: 40.1%	0.7945	0.911	link
AMR	347	T: 53.03%, C: 46.97%	A: 53.03%, G: 46.97%	0.9543	0.9769	link
MXL	64	T: 44.53%, C: 55.47%	A: 45.31%, G: 54.69%	0.9689	1	link
PUR	104	T: 69.23%, C: 30.77%	A: 69.23%, G: 30.77%	0.9118	0.9549	link
CLM	94	T: 61.17%, C: 38.83%	A: 61.17%, G: 38.83%	0.9557	0.9776	link
PEL	85	T: 30.59%, C: 69.41%	A: 30.0%, G: 70.0%	0.9725	1	link
EAS	504	T: 42.66%, C: 57.34%	A: 43.85%, G: 56.15%	0.9213	0.9834	link
CHB	103	T: 51.94%, C: 48.06%	A: 52.43%, G: 47.57%	0.9052	0.9607	link
JPT	104	T: 39.42%, C: 60.58%	A: 40.38%, G: 59.62%	0.9218	0.9795	link
CHS	105	T: 41.9%, C: 58.1%	A: 42.38%, G: 57.62%	0.9424	0.9803	link
CDX	93	T: 34.95%, C: 65.05%	A: 38.17%, G: 61.83%	0.8701	1	link
KHV	99	T: 44.44%, C: 55.56%	A: 45.45%, G: 54.55%	0.96	1	link
EUR	503	T: 67.79%, C: 32.21%	A: 67.59%, G: 32.41%	0.9462	0.9772	link
CEU	99	T: 71.21%, C: 28.79%	A: 71.21%, G: 28.79%	0.9513	0.9754	link
TSI	107	T: 65.42%, C: 34.58%	A: 65.42%, G: 34.58%	0.9591	0.9793	link
FIN	99	T: 60.1%, C: 39.9%	A: 59.09%, G: 40.91%	0.9589	1	link
GBR	91	T: 69.78%, C: 30.22%	A: 68.68%, G: 31.32%	0.9001	0.9735	link
IBS	107	T: 72.43%, C: 27.57%	A: 73.36%, G: 26.64%	0.9538	1	link
SAS	489	T: 38.04%, C: 61.96%	A: 41.92%, G: 58.08%	0.8504	1	link
GIH	103	T: 33.98%, C: 66.02%	A: 40.78%, G: 59.22%	0.7475	1	link
PJL	96	T: 40.62%, C: 59.38%	A: 44.27%, G: 55.73%	0.8613	1	link
BEB	86	T: 35.47%, C: 64.53%	A: 38.37%, G: 61.63%	0.8826	1	link
STU	102	T: 39.22%, C: 60.78%	A: 41.18%, G: 58.82%	0.9217	1	link
ITU	102	T: 40.69%, C: 59.31%	A: 44.61%, G: 55.39%	0.8518	1	link

Showing 1 to 32 of 32 entries

LDpair Tool ⁱ

Investigate correlated alleles for a pair of variants in high LD.

rs26540 rs9326846 7 selected [Calculate](#)

rs9326846
chr5:111301546

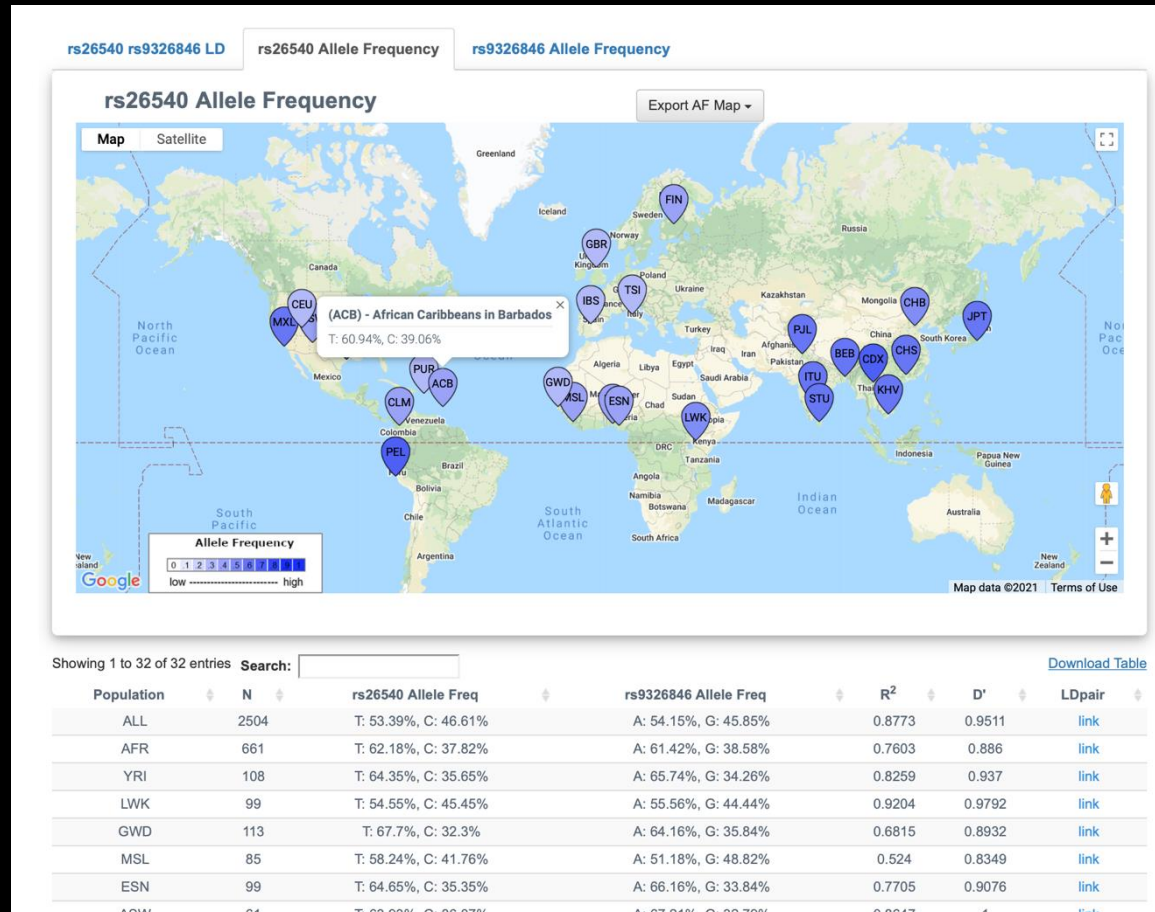
	A	G	
rs26540 C	35	465	500 (0.378)
chr5:111349320 T	777	45	822 (0.622)
	812 (0.614)	510 (0.386)	1322

Haplotypes Statistics

T_A: 777 (0.588) D': 0.886
C_G: 465 (0.352) R²: 0.7603
T_G: 45 (0.034) Chi-sq: 1005.1173
C_A: 35 (0.026) p-value: <0.0001

rs26540(C) allele is correlated with rs9326846(G) allele
rs26540(T) allele is correlated with rs9326846(A) allele

LDlink > LDpop & LDPair



LDlink > LDMatrix

- Input: list of SNPs
- Output: Correlation heatmap



LDlink > LDhap

- Input: list of SNPs
- Output: haplotype frequencies in specified population

LDhap Tool i

Calculate population specific haplotype frequencies of all haplotypes observed for a list of query variants.

rs255930
rs26560

Upload file with variant list

Browse

7 selected

Calculate

RS Number	Position (GRCh37)	Allele Frequencies	Haplotypes				
rs26560	chr5:111338466	G=0.643, C=0.357	G	C	C	G	C
rs255930	chr5:111345538	C=0.685, T=0.315	C	T	C	C	C
rs26542	chr5:111348069	T=0.635, G=0.365	T	G	G	G	T
Haplotype Count			818	411	40	26	21
Haplotype Frequency			0.6188	0.3109	0.0303	0.0197	0.0159

[Download Variant File](#) [Download Haplotype File](#)

LDlink > LDexpress

- Inputs:
 - SNPs
 - window size
 - selected tissues
 - eQTL p-value
 - LD constraints (pop, R^2)
- Output: eQTL associations

LDexpress Tool

Search if a list of variants (or variants in LD with those variants) is associated with gene expression in multiple tissue types.

rs26542

Upload file with variants

Browse

5 selected

All Tissues

R^2

D'

Thresholds

R^2 $\geq$ 0.4

P $<$ 0.05

$\pm$ 100000 base pair window

Calculate

Variants in LD with GTEx QTL

RS Number

rs26542

Show 10 entries

Details for rs26542

Search:

RS Number	Position (GRCh37)	R^2	D'	Gene Symbol	Gencode ID	Tissue	Non-effect Allele Freq	Effect Allele Freq	Effect Size	P-value
rs26542	chr5:111348069	1.0	1.0	EPB41L4A	ENSG00000129595.12	Esophagus - Muscularis	T=0.681	G=0.319	-0.173	6×10^{-5}
rs26542	chr5:111348069	1.0	1.0	EPB41L4A	ENSG00000129595.12	Nerve - Tibial	T=0.681	G=0.319	-0.187	9×10^{-5}
rs26542	chr5:111348069	1.0	1.0	EPB41L4A	ENSG00000129595.12	Thyroid	T=0.681	G=0.319	-0.124	2×10^{-4}
rs255930	chr5:111345538	0.995	1.0	EPB41L4A	ENSG00000129595.12	Esophagus - Muscularis	C=0.68	T=0.32	-0.176	4×10^{-5}
rs255930	chr5:111345538	0.995	1.0	EPB41L4A	ENSG00000129595.12	Nerve - Tibial	C=0.68	T=0.32	-0.186	9×10^{-5}
rs255930	chr5:111345538	0.995	1.0	EPB41L4A	ENSG00000129595.12	Thyroid	C=0.68	T=0.32	-0.130	1×10^{-4}
rs26560	chr5:111338466	0.991	1.0	EPB41L4A	ENSG00000129595.12	Esophagus - Muscularis	G=0.679	C=0.321	-0.175	5×10^{-5}
rs26560	chr5:111338466	0.991	1.0	EPB41L4A	ENSG00000129595.12	Nerve - Tibial	G=0.679	C=0.321	-0.182	1×10^{-4}
rs26560	chr5:111338466	0.991	1.0	EPB41L4A	ENSG00000129595.12	Thyroid	G=0.679	C=0.321	-0.123	2×10^{-4}
rs34942	chr5:111334801	0.986	0.995	EPB41L4A	ENSG00000129595.12	Esophagus - Muscularis	G=0.68	A=0.32	-0.191	1×10^{-5}

Showing 1 to 10 of 303 entries

Previous 1 2 3 4 5 ... 31 Next

[Download GTEx QTL annotated variant list](#)

VEP

Also available as a command-line tool:

<https://useast.ensembl.org/info/docs/tools/vep/script/index.html>

VEP (Variant Effect Predictor)

Required input: List of variants (either paste text or supply a file)
Your choice of transcript database

Variant Effect Predictor ?

New job

Species:

 Homo_sapiens x

Assembly: GRCh38.p13
[Add/remove species](#)
If you are looking for VEP for Human GRCh37, please go to [GRCh37 website](#).

Name for this job (optional):

Input data:

Either paste data:

Examples: [Ensembl default](#), [VCF](#), [Variant identifiers](#), [HGVS notations](#), [SPDI](#)

Or upload file:

Choose File no file selected

Or provide file URL:

Transcript database to use:

☒ Ensembl/GENCODE transcripts

☐ Ensembl/GENCODE basic transcripts

☐ RefSeq transcripts

☐ Ensembl/GENCODE and RefSeq transcripts

VEP

Additional options:

- More transcript identifiers
- Allele frequency and publication data

Additional configurations:

Identifiers Additional identifiers for genes, transcripts and variants

Identifiers

Gene symbol:	☒
Transcript version:	☒
CCDS:	☐
Protein:	☐
UniProt:	☐
HGVS:	☐

Variants and frequency data Co-located variants and frequency data

Variants and frequency data

Find co-located known variants:	Yes
Variant synonyms:	☐
Frequency data for co-located variants:	☒ 1000 Genomes global minor allele frequency ☐ 1000 Genomes continental allele frequencies ☐ ESP allele frequencies ☐ gnomAD (exomes) allele frequencies
PubMed IDs for citations of co-located variants:	☒
Include flagged variants:	☐

VEP

Additional options

- Transcript/protein/regulatory annotation
 - Biotype = e.g., protein-coding, miRNA, pseudogene
 - Which intron/exon is variant in?
 - How close does a variant have to be in order to be "mapped" to a transcript?
 - What protein domain is variant in?
 - Further regulatory consequences
 - Phenotype associations from several more databases

Additional annotations  Additional transcript, protein and regulatory annotations

Transcript annotation

Transcript biotype:



Exon and intron numbers:



Transcript support level:



APPRIS:



MANE:



Identify canonical transcripts:



Upstream/Downstream distance (bp):

miRNA structure:



Protein annotation

Protein domains:



Regulatory data

Get regulatory region consequences:

Phenotype data and citations

Phenotypes:



DisGeNET:



Mastermind:



VEP

Additional options

- SIFT/PolyPhen
(scores for missense variants)
- CADD (combined annotation score
for deleteriousness)
- Splicing/conservation info

Pathogenicity predictions

SIFT:

Prediction and score

PolyPhen:

Prediction and score

dbNSFP:

☒ Disabled

☐ Enabled

CADD:

☐

Condel:

☒ Disabled

☐ Enabled

LoFtool:

☐

Splicing predictions

dbSNV:

☐

MaxEntScan:

☐

SpliceAI:

☐

Conservation

BLOSUM62:

☐

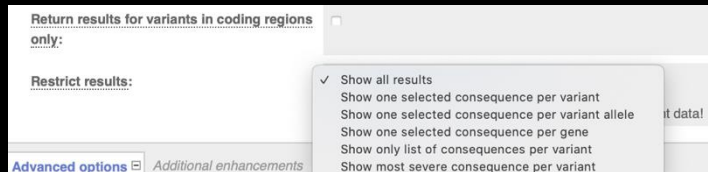
Ancestral allele:

☐

VEP

Additional options

- Filtering by allele freq. (mainly in 1KG)
- Select only certain consequences



Filtering options *Pre-filter results by frequency or consequence type*

Filters

Filter by frequency:

☒ No filtering
☐ Exclude common variants
☐ Advanced filtering

Return results for variants in coding regions only:

☐

Restrict results:

Show all results

NB: Restricting results may exclude biologically important data!

Advanced options *Additional enhancements*

Advanced options

Buffer size:

5000

NB: When the **Regulatory data** option is selected then due to the large the memory requirement but might increase the run time. If you find th

Right align variants prior to consequence calculation:

No

VEP

Click “Run”, wait, and then view the results.
You can access results of multiple runs.

Run >

Recent jobs

Refresh

Show/hide columns (1 hidden)

Filter

Analysis	Jobs	Submitted at
Variant Effect Predictor	VEP analysis of top_snp_proxies in Homo_sapiens Done View results	16/03/2021, 07:36 (GMT)

VEP

You can review the options you used for your run.

The command-line command is also provided for reproducing your annotation.

Variant Effect Predictor results ⓘ

Job details ⓘ

Job summary

Species

Assembly

Options summary

VEP analysis of top_snp_proxies in Homo_sapiens

 Human (Homo sapiens)

GRCh38

1000 Genomes global minor allele frequency:	Enabled
APPRIS:	Enabled
Buffer size:	5000
CADD ^(p) :	Enabled
Filter by frequency:	Disabled
Find co-located known variants:	Enabled
Gene symbol:	Enabled
HGVS:	Enabled
MANE:	Enabled
Phenotypes ^(p) :	Enabled
PolyPhen:	Prediction and score
PubMed IDs for citations of co-located variants:	Enabled
Get regulatory region consequences:	Yes
Restrict results:	Disabled
Right align variants prior to consequence calculation:	Disabled
SIFT:	Prediction and score
Transcript biotype:	Enabled
Transcript database to use:	Ensembl transcripts
Transcript support level:	Enabled
Transcript version:	Enabled
Upstream/Downstream distance (bp):	5000

^(p) = functionality from [VEP plugin](#)

Command line equivalent

```
./vep --af --appris --biotype --buffer_size 500 --check_existing --distance 5000 --hgvs --mane --plugin Phenotypes,dir=[path_to]/,phenotype_feature=1,exclude_sources=COSMIC&HGMD-PUBLIC&Cancer_Gene_Census --p
```

Scroll to see more columns »

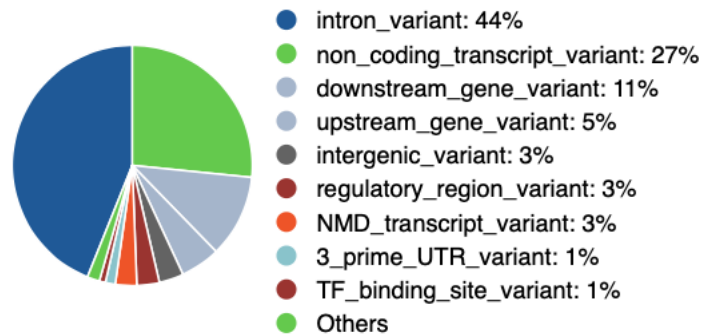
VEP

VEP gives a brief summary of the number of variants provided and their consequences.

Summary statistics

Category	Count
Variants processed	552
Variants filtered out	0
Novel / existing variants	0 (0.0) / 552 (100.0)
Overlapped genes	20
Overlapped transcripts	88
Overlapped regulatory features	47

Consequences (all)



Coding consequences



VEP

Results can be viewed in browser or downloaded as text/VCF.

Results preview

Navigation (per variant)

Page: 1 of 111

Show: 1 5 10 50 All variants

Filters

Uploaded variant

is

defined

Add

Download

All: VCF VEP TXT

BioMart: Variants Genes

New job

Show/hide columns (17 hidden)

Feature type	Feature	Biotype	Exon	HGVSc	cDNA position	Existing variant	Feature strand	MANE SELECT	Transcript support level	APPRIS	AF	Clinical significance	Pubmed	Motif name	Motif position	High info position	Motif score change	TRANSCRIPTION FACTORS	Associated phenotypes	CADD PHRED	CADD RAW
300000060642	Transcript	ENST00000078527.9	protein_coding	-	-	rs71636779	1	-	1	P1	0.0385	-	-	-	-	-	-	-	HYPERPHOSPHATASIA WITH MENTAL RETARDATION SYNDROME 1 (ENSG00000060642.MIM morbid) Hyperphosphatase-intellectual disability syndrome (ENSG00000060642.Orphanet) HYPERPHOSPHATASIA WITH MENTAL RETARDATION (ENSG00000060642.DDG2P)	9.432	0.804097
300000060642	Transcript	ENST00000374145.6	protein_coding	-	-	rs71636779	1	-	2	P1	0.0385	-	-	-	-	-	-	-	HYPERPHOSPHATASIA WITH MENTAL RETARDATION SYNDROME 1 (ENSG00000060642.MIM morbid) Hyperphosphatase-intellectual disability syndrome (ENSG00000060642.Orphanet) HYPERPHOSPHATASIA WITH MENTAL RETARDATION (ENSG00000060642.DDG2P)	9.432	0.804097
300000060642	Transcript	ENST00000431541.6	protein_coding	-	-	rs71636779	1	-	2	-	0.0385	-	-	-	-	-	-	-	Hyperphosphatase-intellectual disability syndrome (ENSG00000060642.Orphanet) HYPERPHOSPHATASIA WITH MENTAL RETARDATION (ENSG00000060642.DDG2P) HYPERPHOSPHATASIA WITH MENTAL RETARDATION SYNDROME 1 (ENSG00000060642.MIM morbid)	9.432	0.804097
300000060642	Transcript	ENST00000455364.2	protein_coding	-	ENST00000455364.2:c.-58+303T>C	rs71636779	1	-	3	P1	0.0385	-	-	-	-	-	-	-	Hyperphosphatase-intellectual disability syndrome (ENSG00000060642.Orphanet) HYPERPHOSPHATASIA WITH MENTAL RETARDATION (ENSG00000060642.DDG2P) HYPERPHOSPHATASIA WITH MENTAL RETARDATION SYNDROME 1 (ENSG00000060642.MIM morbid)	9.432	0.804097
300000060642	Transcript	ENST00000472757.5	nonsense_mediated_decay	-	-	rs71636779	1	-	5	-	0.0385	-	-	-	-	-	-	-	HYPERPHOSPHATASIA WITH MENTAL RETARDATION SYNDROME 1 (ENSG00000060642.MIM morbid) Hyperphosphatase-intellectual disability syndrome (ENSG00000060642.Orphanet) HYPERPHOSPHATASIA WITH MENTAL RETARDATION (ENSG00000060642.DDG2P)	9.432	0.804097
300000060642	Transcript	ENST00000674202.1	protein_coding	-	-	rs71636779	1	NM_017837.4	-	P1	0.0385	-	-	-	-	-	-	-	HYPERPHOSPHATASIA WITH MENTAL RETARDATION (ENSG00000060642.DDG2P) Hyperphosphatase-intellectual disability syndrome (ENSG00000060642.Orphanet) HYPERPHOSPHATASIA WITH MENTAL RETARDATION SYNDROME 1 (ENSG00000060642.MIM morbid)	9.432	0.804097
300000060642	Transcript	ENST00000674222.1	protein_coding	-	-	rs71636779	1	-	-	P1	0.0385	-	-	-	-	-	-	-	Hyperphosphatase-intellectual disability syndrome (ENSG00000060642.Orphanet) HYPERPHOSPHATASIA WITH MENTAL RETARDATION (ENSG00000060642.DDG2P) HYPERPHOSPHATASIA WITH MENTAL RETARDATION SYNDROME 1 (ENSG00000060642.MIM morbid)	9.432	0.804097

VEP

Format the table by selecting columns to show.

Show/hide columns (17 hidden)

☒	Uploaded variant
☒	Location
☒	Allele
☒	Consequence
☐	Impact
☒	Symbol
☒	Gene
☒	Feature type
☒	Feature
☒	Biotype
☒	Exon
☐	Intron
☒	HGVSc
☐	HGVSp
☒	cDNA position
☐	CDS position
☐	Protein position
☐	Amino acids
☐	Codons
☒	Existing variant
☐	Distance to transcript
☒	Feature strand
☐	FLAGS
☐	Symbol source
☐	HGNC ID
☒	MANE SELECT
☐	MANE PLUS CLINICAL
☒	Transcript support level
☒	APPRIS
☐	SIFT
☐	PolyPhen
☐	HGVS offset
☒	AF
☒	Clinical significance
☐	Somatic status
☐	Phenotype or disease
☒	Pubmed
☒	Motif name
☒	Motif position
☒	High info position
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☒	Associated phenotypes
☒	CADD PHRED
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VEP

Download variants Genes

Show/hide columns (31 hidden)													
Uploaded variant	Location	Allele	Consequence	Symbol	Gene	Feature type	Feature	Biotype	Existing variant	Feature strand	AF	CADD PHRED	CADD RAW
rs35755865	1:26813569-26813569	T	upstream_gene_variant	RN7SL165P	ENSG00000241188	Transcript	ENST00000477476.3	misc_RNA	rs35755865	1	0.0078	6.019	Scroll to see more columns »
rs35755865	1:26813569-26813569	T	upstream_gene_variant	AL034380.2	ENSG00000287810	Transcript	ENST00000656287.1	unprocessed_pseudogene	rs35755865	1	0.0078	6.019	
rs35755865	1:26813569-26813569	T	regulatory_region_variant	-	-	RegulatoryFeature	ENSR00000921633	promoter_flanking_region	rs35755865	-	0.0078	6.019	0.454246
rs35583548	1:26813658-26813658	A	upstream_gene_variant	RN7SL165P	ENSG00000241188	Transcript	ENST00000477476.3	misc_RNA	rs35583548	1	0.0527	4.394	0.309076
rs35583548	1:26813658-26813658	A	upstream_gene_variant	AL034380.2	ENSG00000287810	Transcript	ENST00000656287.1	unprocessed_pseudogene	rs35583548	1	0.0527	4.394	0.309076
rs35583548	1:26813658-26813658	A	regulatory_region_variant	-	-	RegulatoryFeature	ENSR00000921633	promoter_flanking_region	rs35583548	-	0.0527	4.394	0.309076
rs35131841	1:26815377-26815377	G	downstream_gene_variant	RN7SL165P	ENSG00000241188	Transcript	ENST00000477476.3	misc_RNA	rs35131841	1	0.0377	12.12	1.040330
rs35131841	1:26815377-26815377	G	upstream_gene_variant	AL034380.2	ENSG00000287810	Transcript	ENST00000656287.1	unprocessed_pseudogene	rs35131841	1	0.0377	12.12	1.040330
rs35131841	1:26815377-26815377	G	regulatory_region_variant	-	-	RegulatoryFeature	ENSR00000921633	promoter_flanking_region	rs35131841	-	0.0377	12.12	1.040330
rs60878021	1:26816063-26816063	C	downstream_gene_variant	RN7SL165P	ENSG00000241188	Transcript	ENST00000477476.3	misc_RNA	rs60878021	1	0.0841	3.198	0.206465
rs60878021	1:26816063-26816063	C	upstream_gene_variant	AL034380.2	ENSG00000287810	Transcript	ENST00000656287.1	unprocessed_pseudogene	rs60878021	1	0.0841	3.198	0.206465
rs60878021	1:26816063-26816063	C	regulatory_region_variant	-	-	RegulatoryFeature	ENSR00000921633	promoter_flanking_region	rs60878021	-	0.0841	3.198	0.206465
rs12728091	1:26817022-26817022	T	downstream_gene_variant	RN7SL165P	ENSG00000241188	Transcript	ENST00000477476.3	misc_RNA	rs12728091	1	0.0397	3.154	0.202683
rs12728091	1:26817022-26817022	T	upstream_gene_variant	AL034380.2	ENSG00000287810	Transcript	ENST00000656287.1	unprocessed_pseudogene	rs12728091	1	0.0397	3.154	0.202683
rs12728091	1:26817022-26817022	T	regulatory_region_variant	-	-	RegulatoryFeature	ENSR00000921633	promoter_flanking_region	rs12728091	-	0.0397	3.154	0.202683

HaploReg

<https://pubs.broadinstitute.org/mammals/haploreg/haploreg.php>

- Documentation

https://pubs.broadinstitute.org/mammals/haploreg/documentation_v4.1.html

- Publication

<https://academic.oup.com/nar/article/40/D1/D930/2903522>

- Background on chromatin states

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3088773/>

HaploReg

- Usage
 - Input formats
 - Comma-delimited list of rsIDs or a single region (e.g., chr3:123456:234567)
 - Text file with one rsID per line
 - OR use the R package HaploR
 - Parameters
 - LD threshold
 - Population for calculating LD (AFR, AMR, ASN, EUR)
 - Source for epigenomes
 - Conservation algorithm
 - Gene set (GENCODE or RefSeq)
 - Display/output options

HaploReg

Set parameters

Build Query **Set Options** Documentation

LD threshold, r^2 (select NA to only show query variants):

1000G Phase 1 population for LD calculation: ☒ AFR ☐ AMR ☐ ASN ☐ EUR

Source for epigenomes:

Mammalian conservation algorithm: ☐ GERP ☒ SiPhy-omega ☐ both

Show position relative to: ☒ GENCODE genes ☐ RefSeq genes ☐ both

Condense lists in table longer than:

Condense indel oligos longer than:

Output mode: ☒ HTML ☐ Text

Select input file and submit (I supplied a list of 10 “lead SNPs” from a fake GWAS)

Build Query **Set Options** Documentation

Use one of the three methods below to enter a set of variants. If an r^2 threshold is specified (see the Set Options tab), results for each variant will be shown in a separate table along with other variants in LD. If r^2 is set to NA, only queried variants will be shown, together in one table.

Query (comma-delimited list of rsIDs OR a single region as chrN:start-end):

or, upload a text file (one refSNP ID per line):

or, select a GWAS:

HaploReg

Output: a table for each SNP

SNP1

SNP2

SNP3

Let's look at these
two →

Query SNP: rs26540 and variants with $r^2 \geq 0.8$																						
chr	pos (hg38)	LD (r ²)	LD (D')	variant	Ref	Alt	AFR freq	AMR freq	ASN freq	EUR freq	SiPhy cons	Promoter histone marks	Enhancer histone marks	DNase	Proteins bound	Motifs changed	NHGRI/EBI GWAS hits	GRASP QTL hits	Selected eQTL hits	GENCODE genes	dbSNP func annot	
5	111976063	0.81	0.92	rs26545	T	G	0.38	0.42	0.55	0.34										NREP-AS1	intronic	
5	111986592	0.81	0.92	rs34933	G	C	0.38	0.42	0.56	0.34						5 altered motifs				NREP-AS1		
5	111990600	0.85	0.96	rs34938	T	C	0.37	0.42	0.55	0.34						4 altered motifs				NREP-AS1		
5	111991576	0.81	0.95	rs40100	C	T	0.38	0.42	0.55	0.34										NREP-AS1		
5	111993295	0.84	0.94	rs26555	A	G	0.38	0.42	0.56	0.34										NREP-AS1		
5	111993952	0.84	0.96	rs26559	C	T	0.37	0.42	0.56	0.34										NREP-AS1		
5	111994444	0.84	0.94	rs34941	A	C	0.38	0.42	0.55	0.34										NREP-AS1		
5	112002769	0.87	0.96	rs26560	G	C	0.37	0.43	0.55	0.34										NREP-AS1		
5	112003403	0.86	0.95	rs27535	T	C	0.38	0.47	0.55	0.38										NREP-AS1		
5	112004636	0.84	0.96	rs25928	A	G	0.37	0.42	0.56	0.33										NREP-AS1		
5	112012372	0.93	1	rs26542	T	G	0.37	0.42	0.56	0.33										NREP-AS1		
5	112012865	0.92	0.97	rs26541	G	C	0.40	0.47	0.55	0.38										NREP-AS1		
5	112013623	1	1	rs26540	T	C	0.39	0.42	0.56	0.33										NREP-AS1		
Query SNP: rs13107325 and variants with $r^2 \geq 0.8$																						
chr	pos (hg38)	LD (r ²)	LD (D')	variant	Ref	Alt	AFR freq	AMR freq	ASN freq	EUR freq	SiPhy cons	Promoter histone marks	Enhancer histone marks	DNase	Proteins bound	Motifs changed	NHGRI/EBI GWAS hits	GRASP QTL hits	Selected eQTL hits	GENCODE genes	dbSNP func annot	
4	102080492	1	1	rs1813006	G	T	0.00	0.03	0.00	0.06						7 altered motifs				5.7kb 3' of BANK1		
4	102113234	1	1	rs13101632	G	A	0.00	0.05	0.00	0.11										38kb 3' of BANK1		
4	102225731	1	1	rs35225200	A	C	0.00	0.05	0.00	0.09						COMP1, RFX5				25kb 3' of SLC39A8		
4	102225733	1	1	rs35518360	A	T	0.00	0.05	0.00	0.09						4 altered motifs				25kb 3' of SLC39A8		
4	102267552	1	1	rs13107325	C	T	0.00	0.04	0.00	0.08						Artd5a, PRDM1, Pax-1		1 hit	7 hits	SLC39A8	missense	
4	102276925	1	1	rs13135092	A	G	0.00	0.05	0.00	0.08						4 altered motifs				SLC39A8	intronic	
Query SNP: rs921919 and variants with $r^2 \geq 0.8$																						
chr	pos (hg38)	LD (r ²)	LD (D')	variant	Ref	Alt	AFR freq	AMR freq	ASN freq	EUR freq	SiPhy cons	Promoter histone marks	Enhancer histone marks	DNase	Proteins bound	Motifs changed	NHGRI/EBI GWAS hits	GRASP QTL hits	Selected eQTL hits	GENCODE genes	dbSNP func annot	
12	124780555	1	1	rs921919	G	A	0.04	0.60	0.37	0.06						6 altered motifs				SCARB1	intronic	
Query SNP: rs13107325 and variants with $r^2 \geq 0.8$																						
chr	pos (hg38)	LD (r ²)	LD (D')	variant	Ref	Alt	AFR freq	AMR freq	ASN freq	EUR freq	SiPhy cons	Promoter histone marks	Enhancer histone marks	DNase	Proteins bound	Motifs changed	NHGRI/EBI GWAS hits	GRASP QTL hits	Selected eQTL hits	GENCODE genes	dbSNP func annot	
4	102080492	1	1	rs1813006	G	T	0.00	0.03	0.00	0.06						7 altered motifs				5.7kb 3' of BANK1		
4	102113234	1	1	rs13101632	G	A	0.00	0.05	0.00	0.11										38kb 3' of BANK1		
4	102225731	1	1	rs35225200	A	C	0.00	0.05	0.00	0.09						COMP1, RFX5				25kb 3' of SLC39A8		
4	102225733	1	1	rs35518360	A	T	0.00	0.05	0.00	0.09						4 altered motifs				25kb 3' of SLC39A8		
4	102267552	1	1	rs13107325	C	T	0.00	0.04	0.00	0.08						Artd5a, PRDM1, Pax-1		1 hit	7 hits	SLC39A8	missense	
4	102276925	1	1	rs13135092	A	G	0.00	0.05	0.00	0.08						4 altered motifs				SLC39A8	intronic	
Query SNP: rs2256609 and variants with $r^2 \geq 0.8$																						
chr	pos (hg38)	LD (r ²)	LD (D')	variant	Ref	Alt	AFR freq	AMR freq	ASN freq	EUR freq	SiPhy cons	Promoter histone marks	Enhancer histone marks	DNase	Proteins bound	Motifs changed	NHGRI/EBI GWAS hits	GRASP QTL hits	Selected eQTL hits	GENCODE genes	dbSNP func annot	
2	21557927	0.8	0.94	rs12168746	C	A	0.04	0.27	0.34	0.16						4 altered motifs				UBE2L3	intronic	
2	21561877	0.81	0.95	rs5754100	T	C	0.04	0.31	0.38	0.18						NF-Y			21 hits	UBE2L3	intronic	
2	21566528	0.95	1	rs58991722	G	C	0.04	0.31	0.40	0.19						ELF1, Ets, Sin3A-20		1 hit	20 hits	UBE2L3	intronic	
2	21568167	0.81	0.95	rs11089620	C	G	0.04	0.32	0.39	0.19						32 tissues	SMC3, YY1			15 altered motifs	UBE2L3	intronic
2	21568615	0.95	1	rs2266959	G	T	0.04	0.31	0.39	0.19						23 tissues	4 bound proteins		1 hit	530 hits	UBE2L3	intronic
2	21570728	1	1	rs2256609	A	G	0.04	0.31	0.40	0.19						19 tissues	BLD, BLD			19 hits	UBE2L3	intronic
2	21574308	0.83	1	rs2266961	C	G	0.05	0.31	0.40	0.19						21 tissues	5 tissues			2 hits	UBE2L3	intronic
2	21575804	0.83	1	rs73166619	C	T	0.05	0.31	0.40	0.19						BLD, PLCNT, GI				20 hits	UBE2L3	intronic
2	21576488	0.83	1	rs5754166	C	T	0.05	0.32	0.39	0.19						BLD				20 hits	UBE2L3	intronic
2	21587718	0.81	0.95	rs9621715	G	A	0.04	0.31	0.39	0.19						19 tissues	4 altered motifs		8 hits	21 hits	UBE2L3	intronic
2	21619030	0.8	0.9	rs4821116	C	T	0.04	0.32	0.39	0.19						6 tissues	BLD	1 hit	6 hits	21 hits	UBE2L3	intronic
Query SNP: rs3749748 and variants with $r^2 \geq 0.8$																						
chr	pos (hg38)	LD (r ²)	LD (D')	variant	Ref	Alt	AFR freq	AMR freq	ASN freq	EUR freq	SiPhy cons	Promoter histone marks	Enhancer histone marks	DNase	Proteins bound	Motifs changed	NHGRI/EBI GWAS hits	GRASP QTL hits	Selected eQTL hits	GENCODE genes	dbSNP func annot	
5	128014857	1	1	rs3749748	C	T	0.03	0.20	0.34	0.21						8 tissues	MUS, GI, SKIN			2 hits	CTC-228N24.3	
5	128021834	1	1	rs17764730	C	T	0.03	0.20	0.34	0.21										6 hits	CTC-228N24.3	
5	128028411	0.84	1	rs34576922	C	G	0.04	0.20	0.34	0.21										24 hits	CTC-228N24.3	intronic
5	128032306	1	1	rs6860245	G	C	0.03	0.20	0.34	0.21										24 hits	CTC-228N24.3	intronic
5	128053152	0.89	1	rs2250127	G	A	0.04	0.19	0.34	0.21						7 tissues				27 hits	CTC-228N24.3	intronic

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Table for **rs13107325** and its LD “friends”

Query SNP: rs13107325 and variants with r ² >= 0.8																					
chr	pos (hg38)	LD (r ²)	LD (D')	variant	Ref	Alt	AFR freq	AMR freq	ASN freq	EUR freq	SiPhy cons	Promoter histone marks	Enhancer histone marks	DNAse	Proteins bound	Motifs changed	NHGR/EBI GWAS hits	GRASP QTL hits	Selected eQTL hits	GENCODE genes	dbSNP func annot
4	102080492	1	1	rs1813006	G	T	0.00	0.03	0.00	0.06						7 altered motifs				5.7kb 3' of BANK1	
4	102113234	1	1	rs13101632	G	A	0.00	0.05	0.00	0.11										38kb 3' of BANK1	
4	102225731	1	1	rs35225200	A	C	0.00	0.05	0.00	0.09						COMP1,RFX5				25kb 3' of SLC39A8	
4	102225733	1	1	rs35518360	A	T	0.00	0.05	0.00	0.09						4 altered motifs	1 hit			25kb 3' of SLC39A8	
4	102267552	1	1	rs13107325	C	T	0.00	0.04	0.00	0.08						Arid5a,PRDM1,Pax-1	7 hits			SLC39A8	missense
4	102276925	1	1	rs13135092	A	G	0.00	0.05	0.00	0.08						4 altered motifs				SLC39A8	intronic

The queried SNP is a missense variant in SLC39A8 with 7 GWAS associations; it alters several motifs and is “constrained.” Nearby genes are given for some of LD “friends.”

Table for **rs2256609** and its LD “friends”

Query SNP: rs2256609 and variants with $r^2 \geq 0.8$																					
chr	pos (hg38)	LD (r ²)	LD (D')	variant	Ref	Alt	AFR freq	AMR freq	ASN freq	EUR freq	SiPhy cons	Promoter histone marks	Enhancer histone marks	DNAse	Proteins bound	Motifs changed	NHGR/EBI GWAS hits	GRASP QTL hits	Selected eQTL hits	GENCODE genes	dbSNP func annot
22	21557927	0.8	0.94	rs12168746	C	A	0.04	0.27	0.34	0.16				BLD		4 altered motifs			17 hits	UBE2L3	intronic
22	21561877	0.81	0.95	rs5754100	T	C	0.04	0.31	0.38	0.18						NF-Y		1 hit	21 hits	UBE2L3	intronic
22	21566528	0.95	1	rs59391722	G	C	0.04	0.31	0.40	0.19			BLD	PANC,BLD		ELF1,Ets,Sin3Ak-20			20 hits	UBE2L3	intronic
22	21568167	0.81	0.95	rs11089620	C	G	0.04	0.32	0.39	0.19		24 tissues		32 tissues	SMC3,YY1	15 altered motifs			20 hits	UBE2L3	intronic
22	21568615	0.95	1	rs2266959	G	T	0.04	0.31	0.39	0.19		23 tissues	SPLN	23 tissues	4 bound proteins	4 altered motifs	1 hit	530 hits	21 hits	UBE2L3	intronic
22	21570728	1	1	rs2256609	A	G	0.04	0.31	0.40	0.19		FAT, BLD	19 tissues	BLD,BLD		4 altered motifs			19 hits	UBE2L3	intronic
22	21574308	0.83	1	rs2266961	C	G	0.05	0.31	0.40	0.19			21 tissues	5 tissues	POL2B	ERalpha-a		2 hits	21 hits	UBE2L3	intronic
22	21575804	0.83	1	rs73166619	C	T	0.05	0.31	0.40	0.19			BLD, PLCNT, GI	ESC,BLD		Foxa			20 hits	UBE2L3	intronic
22	21576488	0.83	1	rs5754166	C	T	0.05	0.32	0.39	0.19			BLD			Mef2			20 hits	UBE2L3	intronic
22	21587718	0.81	0.95	rs9621715	G	A	0.04	0.31	0.39	0.19		BLD, BRN	19 tissues	6 tissues		4 altered motifs		3 hits	21 hits	UBE2L3	intronic
22	21619030	0.8	0.9	rs4821116	C	T	0.04	0.32	0.39	0.19			5 tissues	BLD		Pou2f2	1 hit	6 hits	21 hits	UBE2L3	intronic

The queried SNP is in an intron of UBE2L3. Numerous eQTL associations; histone marks suggest it lies in an enhancer for numerous tissues; DNase sensitivity peaks for “BLD” tissue.

LD “friends” also show protein binding annotation. In the next slide, zoom in on **rs2256609** by clicking the variant name.

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Detail for rs2256609

Detail view for rs2256609

[Link to dbSNP entry](#)

[Link to Ensembl Variation entry](#)

Sequence facts

chr	pos (hg19)	chr	pos (hg38)	Reference	Alternate	1000 Genomes Phase 1 Frequencies				Sequence constraint		dbSNP functional annotation
						AFR	AMR	ASN	EUR	by GERP	by SiPhy	
chr22	21925017	chr22	21570728	A	G	0.04	0.31	0.4	0.19	No	No	intronic

Closest annotated gene

Source	Distance	Direction	ID/Link	Common name	Description
GENCODE	NA	Within gene	ENSG00000185651.10	UBE2L3	ubiquitin-conjugating enzyme E2L 3 [Source:HGNC Symbol;Acc:12488]
RefSeq	NA	Within gene	NR_028437	UBE2L3	

HaploReg

Detail for **rs2256609**

Histone marks for each cell/tissue type

Columns 1-4: cell/tissue identifier

Columns 5-6: chromatin state codes
(see next slide)

Columns 7-11: specific chromatin marks
and DNase sensitivity

Regulatory chromatin states from DNase and histone ChIP-Seq (Roadmap Epigenomics Consortium, 2015)										
(Black = missing data)										
Epigenome ID (EID)	Group	Mnemonic	Description	Chromatin states (Core 15-state model)	Chromatin states (25-state model using 12 imputed marks)	H3K4me1	H3K4me3	H3K27ac	H3K9ac	DNase
E017	IMR90	LNG.IMR90	IMR90 fetal lung fibroblasts Cell Line		12_TxEnhW	H3K4me1_Enh				
E002	ESC	ESC.WA7	ES-WA7 Cells		12_TxEnhW					
E008	ESC	ESC.H9	H9 Cells		12_TxEnhW				H3K9ac_Pro	
E001	ESC	ESC.I3	ES-I3 Cells		12_TxEnhW	H3K4me1_Enh			H3K9ac_Pro	
E015	ESC	ESC.HUES6	HUES6 Cells		12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro			
E014	ESC	ESC.HUES48	HUES48 Cells	7_Enh	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro			
E016	ESC	ESC.HUES64	HUES64 Cells	7_Enh	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E003	ESC	ESC.H1	H1 Cells		12_TxEnhW	H3K4me1_Enh			H3K9ac_Pro	
E024	ESC	ESC.4STAR	ES-UCSF4 Cells		12_TxEnhW	H3K4me1_Enh				
E020	IPSC	IPSC.20B	IPS-20b Cells	7_Enh	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh	H3K9ac_Pro	
E019	IPSC	IPSC.18	IPS-18 Cells		12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro		H3K9ac_Pro	
E018	IPSC	IPSC.15b	IPS-15b Cells	7_Enh	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro		H3K9ac_Pro	
E021	IPSC	IPSC.DF.6.9	IPS DF 6.9 Cells		12_TxEnhW					
E022	IPSC	IPSC.DF.19.11	IPS DF 19.11 Cells							
E007	ES-deriv	ESDR.H1.NEUR.PROG	H1 Derived Neuronal Progenitor Cultured Cells		12_TxEnhW					
E009	ES-deriv	ESDR.H9.NEUR.PROG	H9 Derived Neuronal Progenitor Cultured Cells		12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro			
E010	ES-deriv	ESDR.H9.NEUR	H9 Derived Neuron Cultured Cells	6_EnhG	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro			
E013	ES-deriv	ESDR.CD56.MESO	hESC Derived CD56+ Mesoderm Cultured Cells	6_EnhG	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E012	ES-deriv	ESDR.CD56.ECTO	hESC Derived CD56+ Ectoderm Cultured Cells	7_Enh	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro			
E011	ES-deriv	ESDR.CD184.ENDO	hESC Derived CD184+ Endoderm Cultured Cells	9_TxFink	9_TxReg	H3K4me1_Enh	H3K4me3_Pro		H3K9ac_Pro	
E004	ES-deriv	ESDR.H1.BMP4.MESO	H1 BMP4 Derived Mesoderm Cultured Cells	7_Enh	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro		H3K9ac_Pro	
E005	ES-deriv	ESDR.H1.BMP4.TROP	H1 BMP4 Derived Trophoblast Cultured Cells	7_Enh	12_TxEnhW	H3K4me1_Enh				
E006	ES-deriv	ESDR.H1.MSC	H1 Derived Mesenchymal Stem Cells	7_Enh	12_TxEnhW					
E062	Blood & T-cell	BLD.PER.MONUC.PC	Primary mononuclear cells from peripheral blood	7_Enh	12_TxEnhW	H3K4me1_Enh			H3K9ac_Pro	
E034	Blood & T-cell	BLD.CD3.PPC	Primary T cells from peripheral blood	7_Enh	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		DNase
E045	Blood & T-cell	BLD.CD4.CD25L.CD127.TMEMPC	Primary T cells effector/memory enriched from peripheral blood	7_Enh	10_TxEnh5	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E033	Blood & T-cell	BLD.CD3.CPC	Primary T cells from cord blood	6_EnhG	10_TxEnh5	H3K4me1_Enh	H3K4me3_Pro			
E044	Blood & T-cell	BLD.CD4.CD25.CD127M.TREGPC	Primary T regulatory cells from peripheral blood	7_Enh	10_TxEnh5	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E043	Blood & T-cell	BLD.CD4.CD25M.TPC	Primary T helper cells from peripheral blood	7_Enh	10_TxEnh5	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E039	Blood & T-cell	BLD.CD4.CD25M.CD45RA.NPC	Primary T helper naive cells from peripheral blood	7_Enh	10_TxEnh5	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E041	Blood & T-cell	BLD.CD4.CD25M.IL17M.PL.TPC	Primary T helper cells PMA-1 stimulated	7_Enh	9_TxReg	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E042	Blood & T-cell	BLD.CD4.CD25M.IL17P.PL.TPC	Primary T helper 17 cells PMA-1 stimulated	7_Enh	10_TxEnh5	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E040	Blood & T-cell	BLD.CD4.CD25M.CD45RO.MPC	Primary T helper memory cells from peripheral blood 1	7_Enh	9_TxReg	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E037	Blood & T-cell	BLD.CD4.MPC	Primary T helper memory cells from peripheral blood 2	7_Enh	9_TxReg	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E048	Blood & T-cell	BLD.CD8.MPC	Primary T CD8+ memory cells from peripheral blood	7_Enh	10_TxEnh5	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E038	Blood & T-cell	BLD.CD4.NPC	Primary T helper naive cells from peripheral blood	7_Enh	10_TxEnh5	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh	H3K9ac_Pro	
E047	Blood & T-cell	BLD.CD8.NPC	Primary T CD8+ naive cells from peripheral blood	7_Enh	10_TxEnh5	H3K4me1_Enh	H3K4me3_Pro		H3K9ac_Pro	
E029	HSC & B-cell	BLD.CD14.PC	Primary monocytes from peripheral blood	7_Enh	10_TxEnh5	H3K4me1_Enh		H3K27ac_Enh		
E031	HSC & B-cell	BLD.CD19.CPC	Primary B cells from cord blood	7_Enh	10_TxEnh5	H3K4me1_Enh				
E035	HSC & B-cell	BLD.CD34.PC	Primary hematopoietic stem cells	7_Enh	9_TxReg	H3K4me1_Enh				
E051	HSC & B-cell	BLD.MOB.CD34.PC.M	Primary hematopoietic stem cells G-CSF-mobilized Male	2_TssAFink	9_TxReg	H3K4me1_Enh	H3K4me3_Pro			
E050	HSC & B-cell	BLD.MOB.CD34.PC.F	Primary hematopoietic stem cells G-CSF-mobilized Female	7_Enh	9_TxReg	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E036	HSC & B-cell	BLD.CD34.CC	Primary hematopoietic stem cells short term culture	2_TssAFink	9_TxReg	H3K4me1_Enh	H3K4me3_Pro			
E032	HSC & B-cell	BLD.CD19.PPC	Primary B cells from peripheral blood	7_Enh	12_TxEnhW	H3K4me1_Enh		H3K27ac_Enh		
E046	HSC & B-cell	BLD.CD56.PC	Primary Natural Killer cells from peripheral blood	7_Enh	12_TxEnhW	H3K4me1_Enh		H3K27ac_Enh		
E030	HSC & B-cell	BLD.CD15.PC	Primary neutrophils from peripheral blood	7_Enh	9_TxReg	H3K4me1_Enh	H3K4me3_Pro			
E026	Mesench	STRM.MRW.MSC	Bone Marrow Derived Cultured Mesenchymal Stem Cells	7_Enh	10_TxEnh5	H3K4me1_Enh	H3K4me3_Pro		H3K9ac_Pro	
E049	Mesench	STRM.CHON.MRW.DR.MSC	Mesenchymal Stem Cell Derived Chondrocyte Cultured Cells	7_Enh	9_TxReg	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh	H3K9ac_Pro	
E025	Mesench	FAT.ADIP.DR.MSC	Adipose Derived Mesenchymal Stem Cell Cultured Cells	7_Enh	9_TxReg	H3K4me1_Enh	H3K4me3_Pro		H3K9ac_Pro	
E023	Mesench	FAT.MSC.DR.ADIP	Mesenchymal Stem Cell Derived Adipocyte Cultured Cells	2_TssAFink	9_TxReg	H3K4me1_Enh	H3K4me3_Pro		H3K9ac_Pro	
E052	Myosat	MUS.SAT	Muscle Satellite Cultured Cells	7_Enh	10_TxEnh5	H3K4me1_Enh	H3K4me3_Pro		H3K9ac_Pro	
E055	Epithelial	SKIN.PEN.FRSK.FIB.01	Foreskin Fibroblast Primary Cells skin01		12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E056	Epithelial	SKIN.PEN.FRSK.FIB.02	Foreskin Fibroblast Primary Cells skin02		12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E059	Epithelial	SKIN.PEN.FRSK.MEL.01	Foreskin Melanocyte Primary Cells skin01	7_Enh	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro			
E061	Epithelial	SKIN.PEN.FRSK.MEL.03	Foreskin Melanocyte Primary Cells skin03	7_Enh	10_TxEnh5	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E057	Epithelial	SKIN.PEN.FRSK.KER.02	Foreskin Keratinocyte Primary Cells skin02	6_EnhG	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro			
E058	Epithelial	SKIN.PEN.FRSK.KER.03	Foreskin Keratinocyte Primary Cells skin03	3_TxFink	9_TxReg	H3K4me1_Enh	H3K4me3_Pro	H3K27ac_Enh		
E028	Epithelial	BRST.HMEC.35	Breast variant Human Mammary Epithelial Cells (vHMEC)	7_Enh	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro			
E027	Epithelial	BRST.MYO	Breast Myoepithelial Primary Cells	7_Enh	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro		H3K9ac_Pro	
E054	Neurosph	BRN.GANGEM.DR.NRSPHR	Ganglion Eminence derived primary cultured neurospheres	7_Enh	12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro			
E053	Neurosph	BRN.CRTX.DR.NRSPHR	Cortex derived primary cultured neurospheres	7_Enh	12_TxEnhW	H3K4me1_Enh				
E112	Thymus	THYM	Thymus		12_TxEnhW	H3K4me1_Enh	H3K4me3_Pro			

HaploReg

Key to coloring/abbreviations in the chromatin states columns

The states are as follows

STATE NO.	MNEMONIC	DESCRIPTION	COLOR NAME	COLOR CODE
1	TssA	Active TSS	Red	255,0,0
2	TssAFlnk	Flanking Active TSS	Orange Red	255,69,0
3	TxFlnk	Transcr. at gene 5' and 3'	LimeGreen	50,205,50
4	Tx	Strong transcription	Green	0,128,0
5	TxWk	Weak transcription	DarkGreen	0,100,0
6	EnhG	Genic enhancers	GreenYellow	194,225,5
7	Enh	Enhancers	Yellow	255,255,0
8	ZNF/Rpts	ZNF genes & repeats	Medium Aquamarine	102,205,170
9	Het	Heterochromatin	PaleTurquoise	138,145,208
10	TssBiv	Bivalent/Poised TSS	IndianRed	205,92,92
11	BivFlnk	Flanking Bivalent TSS/Enh	DarkSalmon	233,150,122
12	EnhBiv	Bivalent Enhancer	DarkKhaki	189,183,107
13	ReprPC	Repressed PolyComb	Silver	128,128,128
14	ReprPCWk	Weak Repressed PolyComb	Gainsboro	192,192,192
15	Quies	Quiescent/Low	White	255,255,255

The states are as follows

STATE NO.	MNEMONIC	DESCRIPTION	COLOR NAME	COLOR CODE
1	TssA	Active TSS	Red	255,0,0
2	PromU	Promoter Upstream TSS	Orange Red	255,69,0
3	PromD1	Promoter Downstream TSS 1	Orange Red	255,69,0
4	PromD2	Promoter Downstream TSS 2	Orange Red	255,69,0
5	Tx5	Transcribed - 5' preferential	Green	0,128,0
6	Tx	Strong transcription	Green	0,128,0
7	Tx3	Transcribed - 3' preferential	Green	0,128,0
8	TxWk	Weak transcription	Lighter Green	0,150,0
9	TxReg	Transcribed & regulatory (Prom/Enh)	Electric Lime	194,225,5
10	TxEnh5	Transcribed 5' preferential and Enh	Electric Lime	194,225,5
11	TxEnh3	Transcribed 3' preferential and Enh	Electric Lime	194,225,5
12	TxEnhW	Transcribed and Weak Enhancer	Electric Lime	194,225,5
13	EnhA1	Active Enhancer 1	Orange	255,195,77
14	EnhA2	Active Enhancer 2	Orange	255,195,77
15	EnhAF	Active Enhancer Flank	Orange	255,195,77
16	EnhW1	Weak Enhancer 1	Yellow	255,255,0
17	EnhW2	Weak Enhancer 2	Yellow	255,255,0
18	EnhAc	Primary H3K27ac possible Enhancer	Yellow	255,255,0
19	DNase	Primary DNase	Lemon	255,255,102
20	ZNF/Rpts	ZNF genes & repeats	Aquamarine	102,205,170
21	Het	Heterochromatin	Light Purple	138,145,208
22	PromP	Poised Promoter	Pink	230,184,183
23	PromBiv	Bivalent Promoter	Dark Purple	112,48,160
24	ReprPC	Repressed Polycomb	Gray	128,128,128
25	Quies	Quiescent/Low	White	255,255,255

RegulomeDB

<https://regulomedb.org>

- Usage

- Input formats (text box only)
 - Enter rsIDs line by line
 - Or enter coordinate ranges line by line (chr6:12345:23456)
 - OR use the R package HaploR
- No parameters

Documentation:

<https://regulomedb.org/regulome-help/>

Reference:

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3431494/>

RegulomeDB

Searching a list of SNPs



🔍 Search by dbSNP ID or coordinate range (hg19)

rs13107325
rs921919
rs13107325
rs2256609
rs3749748
rs72801474
rs2502465
rs9604045
rs12748152

*Click for example entry: **multiple dbSNPs** or **coordinates ranges***

Search

Use RegulomeDB to identify DNA features and regulatory elements in non-coding regions of the human genome by entering ...

- ▶ DBSNP IDS
- ▶ SINGLE NUCLEOTIDES
- ▶ A CHROMOSOMAL REGION

RegulomeDB

Output – click on a SNP

▼ How to interpret the new RegulomeDB probability score?

The RegulomeDB probability score is ranging from 0 to 1, with 1 being most likely to be a regulatory variant. The RegulomeDB score represents a model integrating functional genomics features along with continuous values such as ChIP-seq signal, DNase-seq signal, information content change, and DeepSEA scores among others. For more detail see our manuscript:

Dong S and Boyle AP. Predicting functional variants in enhancer and promoter elements using RegulomeDB. Human Mutation 2019, 40:1292-1298. PMID: 31228310.

There is an overall positive correlation between the ranking scores and the probability scores, but there are some exceptions because 1) we added additional features when predicting probability scores. 2) features used in probability scoring were weighted differently from ranking scoring. We are working on further refining our algorithm.



This search has found **9** variant(s).

Download BED

Download TSV

Chromosome location	dbSNP IDs	Rank	Score
chr1:27138392..27138393	rs12748152	4	0.60906
chr4:103188708..103188709	rs13107325	5	0.17627
chr5:111349319..111349320	rs26540	5	0.48681
chr5:127350548..127350549	rs3749748	4	0.60906
chr5:132444127..132444128	rs72801474	4	0.60906
chr6:75455384..75455385	rs2502465	6	0.13144
chr12:125265200..125265201	rs921919	2b	0.63769
chr13:113927207..113927208	rs9604045	4	0.60906
chr22:21925016..21925017	rs2256609	5	0.6855

Let's click on rs921919 (lowest "rank")

▼ What does the RegulomeDB rank represent?

The scoring scheme refers to the following available datatypes for a single coordinate.

Score Supporting data

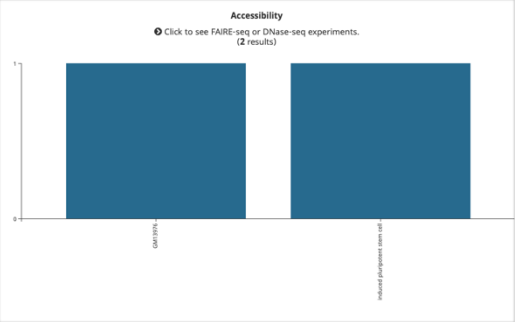
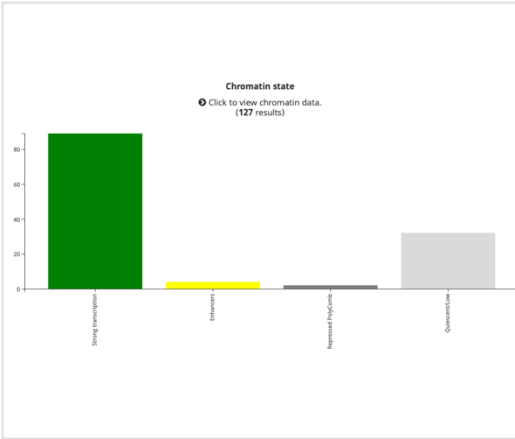
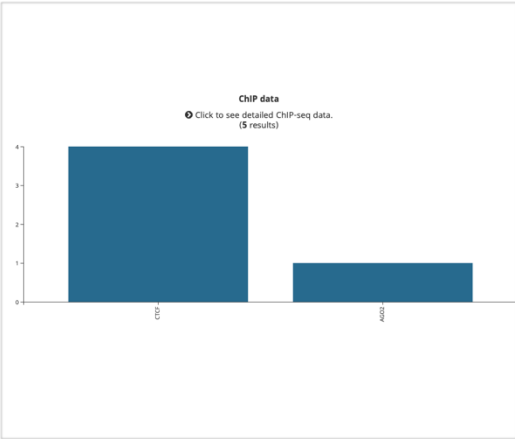
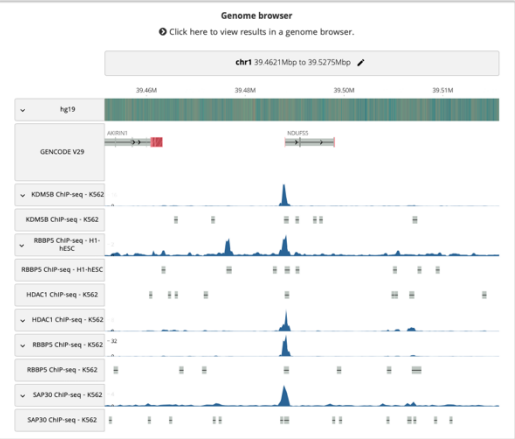
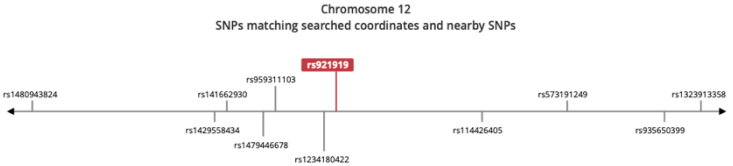
- | | |
|----|---|
| 1a | eQTL + TF binding + matched TF motif + matched DNase Footprint + DNase peak |
| 1b | eQTL + TF binding + any motif + DNase Footprint + DNase peak |
| 1c | eQTL + TF binding + matched TF motif + DNase peak |
| 1d | eQTL + TF binding + any motif + DNase peak |
| 1e | eQTL + TF binding + matched TF motif |
| 1f | eQTL + TF binding / DNase peak |
| 2a | TF binding + matched TF motif + matched DNase Footprint + DNase peak |
| 2b | TF binding + any motif + DNase Footprint + DNase peak |
| 2c | TF binding + matched TF motif + DNase peak |
| 3a | TF binding + any motif + DNase peak |
| 3b | TF binding + matched TF motif |
| 4 | TF binding + DNase peak |
| 5 | TF binding or DNase peak |
| 6 | Motif hit |
| 7 | Other |

RegulomeDB

Detail for rs921919

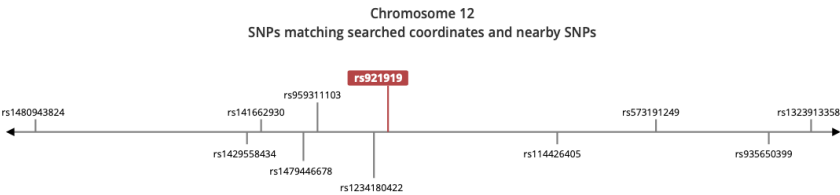


SEARCHED COORDINATES	chr12:125265200-125265201
PEAKS	11 peaks
RANK	2b
SCORE	0.63769
RS921919	G=0.5185, A=0.4815 (GnomAD) G=0.6052, A=0.3948 (1000Genomes) G=0.5292, A=0.4708 (TOPMED) G=0.375, A=0.625 (NorthernSweden) G=0.3277, A=0.6723 (ALSPAC) G=0.3182, A=0.6818 (TWINSUK) G=0.6274, A=0.3726 (Vietnamese) G=0.3502, A=0.6498 (Estonian) 5 less



RegulomeDB

Detail for rs921919



Genome browser

ChIP data

Chromatin state

Accessibility

Motifs

QTL data

CTCF

Method ^	Peak location ⇅	Biosample ⇅	Targets ⇅	Organ ⇅	Dataset ⇅	File ⇅	Value ⇅	Strand ⇅
ChIP-seq	chr12:125264956..125265206	SK-N-SH	CTCF	brain	ENCSR000EIC	ENCFF861DPF	27.04784	⬅
ChIP-seq	chr12:125264886..125265296	OCI-LY3	CTCF	bone element	ENCSR756ZKG	ENCFF653ZMJ	57.59350	⬅
ChIP-seq	chr12:125265174..125265838	HepG2	AGO2	liver	ENCSR151NQL	ENCFF622VSV	41.25560	⬅
ChIP-seq	chr12:125264839..125265339	A549	CTCF	lung	ENCSR000AUE	ENCFF335GSE	18.03911	⬅
ChIP-seq	chr12:125264977..125265233	retinal pigment epithelial cell	CTCF	eye, epithelium	ENCSR000DVI	ENCFF139DOR	11.18871	⬅

Genome browser

ChIP data

Chromatin state

Accessibility

Motifs

QTL data

FAIRE-seq and DNase-seq experiments

Q

Search for a biosample name

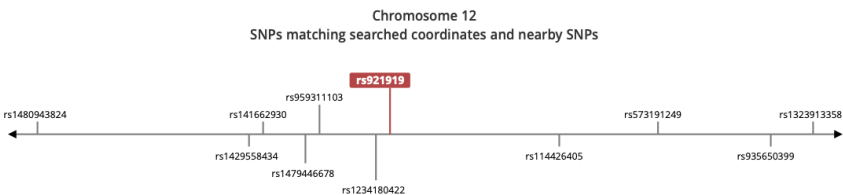
x

GM13976 1

File	Dataset	Organ	Method
ENCFF001UVF	ENCSR000EJG	lymph node	DNase-seq

induced pluripotent stem cell 1

File	Dataset	Organ	Method
ENCFF001UWG	ENCSR000EKH		DNase-seq



Genome browser

ChIP data

Chromatin state

Accessibility

Motifs

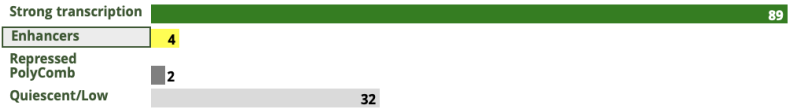
QTL data

Chromatin state

Q

Search for a biosample name, chromatin state, or organ

x



Chromatin state ^	Chromatin state window ⇅	Biosample ⇅	Organ ⇅	Dataset ⇅	File ⇅
Enhancers	chr12:125261800..125271200	adrenal gland	adrenal gland	ENCSR477KTS	TSTFF154468
Enhancers	chr12:125263600..125266000	muscle of trunk	musculature of body	ENCSR152KQC	TSTFF196498
Enhancers	chr12:125258600..125267600	myoepithelial cell of mammary gland	epithelium, mammary gland	ENCSR824PCE	TSTFF800715
Enhancers	chr12:125263400..125266000	GM12878	blood, bodily fluid	ENCSR538BFF	TSTFF941751

RegulomeDB

Detail for rs921919



GWAS Catalog

<https://www.ebi.ac.uk/gwas/>

Input for “direct access”

- SNP/variant
- Region
- Gene
- Phenotype
- Author/study

Some forms of output

- Variant
- Region
- Gene
- Trait
- Publication



GWAS Catalog

The NHGRI-EBI Catalog of human genome-wide association studies

Search the catalog



Examples: [breast carcinoma](#), [rs7329174](#), [Yao](#), [2q37.1](#), [HBS1L](#), [6:16000000-25000000](#)

GWAS Catalog

Examples of variant and gene “tags”

Refine search results

V Variants 1

G Genes 1

Catalog stats

- Last data release on 2021-02-25
- 4892 publications
- 154381 SNPs
- 248356 associations
- Genome assembly GRCh38.p13
- dbSNP Build 153
- Ensembl Build 100

Search results for *rs7329174*

V **rs7329174**

Location: 13:40983974 Cytogenetic region:13q14.11 Most severe consequence: Intron variant Mapped gene(s): ELF1

Associations 4 Studies 4

G **ELF1**

Description: E74 like ETS transcription factor 1

Location: 13:40931924-41061440 Cytogenetic region: 13q14.11 Biotype: protein coding

Associations 27 Studies 27

GWAS Catalog

Variant → Tables of associations, studies, and traits

Variant information

Location ⓘ

Cytogenetic region ⓘ

Most severe consequence ⓘ

Mapped gene(s) ⓘ

13:40983974

13q14.11

Intron variant

ELF1

Alleles ⓘ

Minor allele ⓘ

MAF ⓘ

A/G (forward strand)

G

0.1072

Variant in Ensembl ⓘ

• Genomic context ⓘ

• Phenotype information ⓘ

• Genes and regulation ⓘ

• Population genetics ⓘ

• Citations ⓘ

Variant in dbSNP ⓘ

Variant in UCSC ⓘ

ⓘ GWAS Catalog data is currently mapped to Genome Assembly GRCh38.p13 and dbSNP Build153

Available data:

Associations 4

Studies 4

Traits 2

Linkage disequilibrium (LD)

Download Catalog data ⓘ

Associations 4

Search

🔍

⌵

Variant and risk allele	P-value	P-value annotation	RAF	OR	Beta	CI	Mapped gene	Reported trait	Trait(s)	Study accession	Location
rs7329174-G	8 x 10 ⁻⁹		NR	1.27	-	[1.17-1.38]	ELF1	Crohn's disease	Crohn's disease	GCST001785	13:40983974
rs7329174-G	1 x 10 ⁻⁸		0.22	1.26	-	[1.16-1.36]	ELF1	Systemic lupus erythematosus	systemic lupus erythematosus	GCST000858	13:40983974
rs7329174-G	6 x 10 ⁻⁶		0.21	1.45	-	-	ELF1	Systemic lupus erythematosus	systemic lupus erythematosus	GCST001795	13:40983974
rs7329174-T	3 x 10 ⁻⁶	(Chinese)	0.211	1.27	-	[NR]	ELF1	Systemic lupus erythematosus	systemic lupus erythematosus	GCST003622	13:40983974

Studies 4

Search

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⌵

First author	Study accession	Publication date ⓘ	Journal	Title	Reported trait	Trait(s)	Discovery sample number and ancestry	Replication sample number and ancestry	Association count	Summary statistics
Yang J	GCST000858	2010-11-02	Hum Mol Genet	ELF1 is associated with systemic lupus erythematosus in Asian populations.	Systemic lupus erythematosus	systemic lupus erythematosus	• 1772 East Asian	• 4071 East Asian • 1413 South East Asian	1	NA
Yamazaki K	GCST001785	2012-12-21	Gastroenterology	A genome-wide association study identifies 2 susceptibility loci for Crohn's disease in a Japanese population.	Crohn's disease	Crohn's disease	• 3761 East Asian	• 16951 East Asian	11	NA
Yang W	GCST001795	2012-12-27	Am J Hum Genet	Meta-analysis followed by replication identifies loci in or near CDKN1B, TET3, CD80, DRAM1, and ARID5B as associated with systemic lupus erythematosus in Asians.	Systemic lupus erythematosus	systemic lupus erythematosus	• 5050 East Asian	• 8923 East Asian • 1416 South East Asian	34	NA
Morris DL	GCST003622	2016-07-11	Nat Genet	Genome-wide association meta-analysis in Chinese and European individuals identifies ten new loci associated with systemic lupus erythematosus.	Systemic lupus erythematosus	systemic lupus erythematosus	• 5057 East Asian • 10995 European	• 11675 European • 8117 East Asian	74	NA

Traits 2

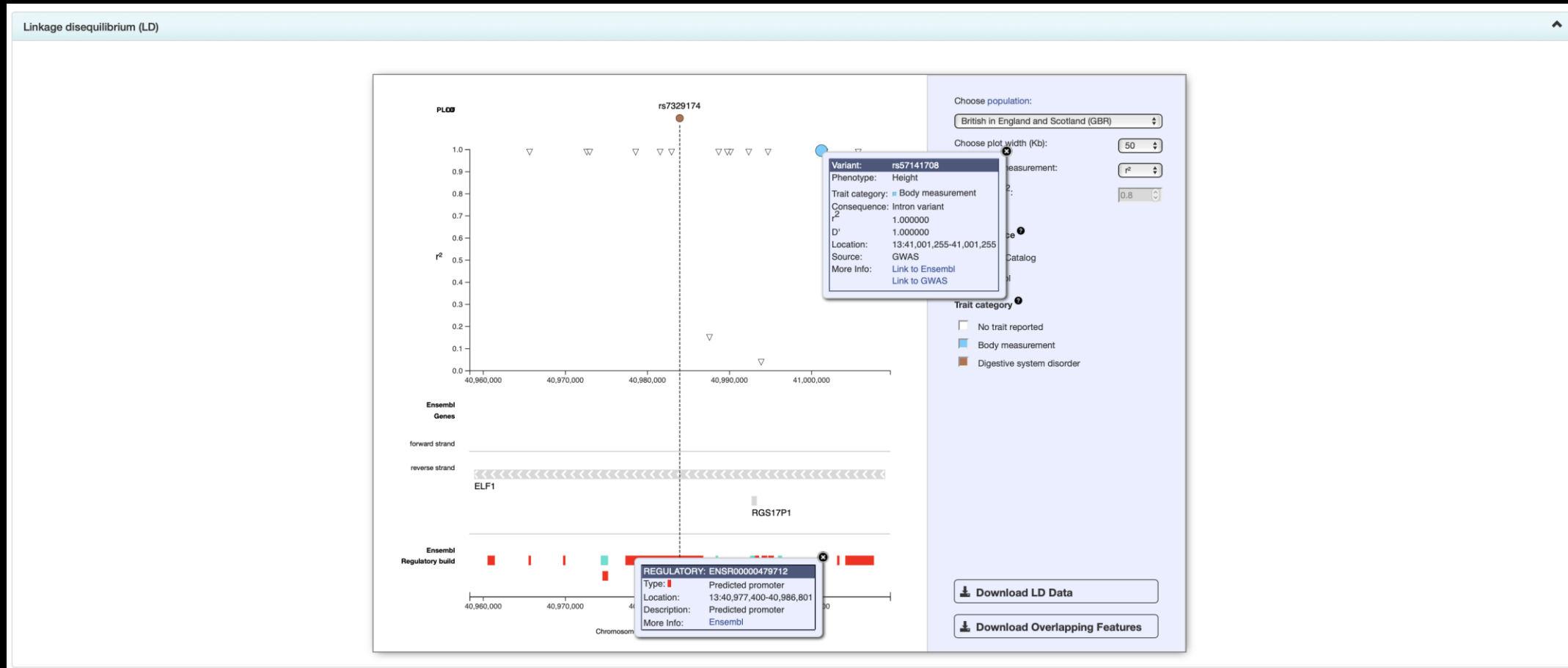
Search

🔍

⌵

GWAS Catalog

Variant → LD plot with some tracks



GWAS Catalog

Gene → Tables of
associations, studies,
and traits

Gene information

Gene name

BCAS4

Description

breast carcinoma amplified sequence 4

Location

20:50794894-50882676

Cytogenetic region

20q13.13

Biotype

protein_coding

Gene in NCBI

Gene in Ensembl

Phenotype information

Pathway information

Regulation

Expression

Gene in Open Targets

Gene in OMIM

Available data:

Associations11

Studies3

Traits9

Download Catalog data

Associations11

Search

Variant and risk allele	P-value	P-value annotation	RAF	OR	Beta	CI	Mapped gene	Reported trait	Trait(s)	Study accession	Location
rs6020772-T	1 x 10 ⁻⁶	(EA)	0.2491	-	0.0348 unit decrease	[0.021-0.049]	BCAS4	Breast cancer	breast carcinoma	GCST004988	20:50826118
rs6013012-T	1 x 10 ⁻⁷		NR	1.12342	-	[1.07596-1.17298]	BCAS4, PARD6B	Non-melanoma skin cancer	non-melanoma skin carcinoma	GCST005896	20:50775369
rs6122972-A	3 x 10 ⁻⁶		0.82	2.02	-	[1.46-2.80]	PARD6B, BCAS4	Bipolar disorder	bipolar disorder	GCST001164	20:50758705
rs62202837-C	5 x 10 ⁻¹⁷		0.769557	1.0869565	-	[1.06-1.11]	BCAS4, PARD6B	Basal cell carcinoma	basal cell carcinoma	GCST008871	20:50782470
rs62202837-C	2 x 10 ⁻¹³		0.758185	1.0752687	-	[1.05-1.1]	BCAS4, PARD6B	Keratinocyte cancer (MTAG)	keratinocyte carcinoma	GCST008870	20:50782470

Showing 1 to 5 of 11 rows5 rows per page

Studies3

Search

First author	Study accession	Publication date	Journal	Title	Reported trait	Trait(s)	Discovery sample number and ancestry	Replication sample number and ancestry	Association count	Summary statistics
Yosifova A	GCST001164	2011-07-19	Genes Brain Behav	Genome-wide association study on bipolar disorder in the Bulgarian population.	Bipolar disorder	bipolar disorder	• 564 Other	• 450 Other	3	NA
Liyanage UE	GCST008871	2019-06-07	Hum Mol Genet	Combined analysis of keratinocyte cancers identifies novel genome-wide loci.	Basal cell carcinoma	basal cell carcinoma	• 651138 European	-	76	NA
Visconti A	GCST005896	2018-05-08	Nat Commun	Genome-wide association study in 176,678 Europeans reveals genetic loci for tanning response to sun exposure.	Non-melanoma skin cancer	non-melanoma skin carcinoma	• 187652 NR	-	82	NA
Liyanage UE	GCST008870	2019-06-07	Hum Mol Genet	Combined analysis of keratinocyte cancers identifies novel genome-wide loci.	Keratinocyte cancer (MTAG)	keratinocyte carcinoma	• 358840 European	-	88	NA
Law PJ	GCST007992	2019-05-14	Nat Commun	Association analyses identify 31 new risk loci for colorectal cancer susceptibility.	Colorectal cancer	colorectal cancer	• 92967 European	-	87	NA

Traits 9		
Search		
Trait label(s)	Reported trait(s)	Association count with BCAS4
PHF-tau measurement	Total PHF-tau (SNP x SNP interaction)	3
breast carcinoma	Breast cancer	1
non-melanoma skin carcinoma	Non-melanoma skin cancer	1
bipolar disorder	Bipolar disorder	1
basal cell carcinoma	Basal cell carcinoma	1
Showing 1 to 5 of 9 rows 5 rows per page		

GWAS Catalog

Region → Tables of
associations, studies,
and traits

Region information

Region name

2q37.1

Region in NCBI

Location

2:230100001-234700000

Region in Ensembl

Available data:

Associations613

Studies351

Traits201

Download Catalog data

Associations613

Search

Variant and risk allele	P-value	P-value annotation	RAF	OR	Beta	CI	Mapped gene	Reported trait	Trait(s)	Study accession	Location
rs10185460-T	4 x 10 ⁻¹¹		0.328309	-	0.0214139 unit decrease	[0.015-0.028]	NPPC, DIS3L2	Male-pattern baldness	balding measurement	GCST007020	2:231939812
rs4973576-T	1 x 10 ⁻⁹		-	-	0.0145232 unit decrease	[0.0099-0.0192]	NGEF	Leisure sedentary behaviour (television watching)	lifestyle measurement	GCST010084	2:232942061
rs146859528-T	3 x 10 ⁻⁹		-	-	0.25 unit increase	[0.17-0.33]	CHRNA	Spherical equivalent	refractive error measurement	GCST010378	2:232541117
rs1550094-A	2 x 10 ⁻⁶¹		-	-	0.19 unit increase	[0.17-0.21]	PRSS56	Spherical equivalent	refractive error measurement	GCST010378	2:232520686
rs1190463-A	2 x 10 ⁻¹¹		-	-	0.1 unit increase	[0.08-0.12]	PRSS56, ECEL1	Spherical equivalent	refractive error measurement	GCST010378	2:232495341

Showing 1 to 5 of 613 rows5 rows per page

12345...123

Studies351

Search

First author	Study accession	Publication date	Journal	Title	Reported trait	Trait(s)	Discovery sample number and ancestry	Replication sample number and ancestry	Association count	Summary statistics
Cox AJ	GCST001976	2013-04-19	Atherosclerosis	Association of SNPs in the UGT1A gene cluster with total bilirubin and mortality in the Diabetes Heart Study.	Bilirubin levels	bilirubin measurement	• 1180 European	-	1	NA
McGue M	GCST002135	2013-08-14	Behav Genet	A genome-wide association study of behavioral disinhibition.	Nicotine use	nicotine use	• 7188 European	-	6	NA
McGue M	GCST002130	2013-08-14	Behav Genet	A genome-wide association study of behavioral disinhibition.	Alcohol consumption	alcohol drinking	• 7188 European	-	2	NA
Seo JY	GCST010048	2020-02-21	PLoS One	A genome-wide association study on liver enzymes in Korean population.	Bilirubin levels	bilirubin measurement	• 4457 East Asian	• 2031 East Asian	4	NA
Vaidyanathan U	GCST002710	2014-12-01	Psychophysiology	Heritability and molecular genetic basis of antisaccade eye tracking error rate: a genome-wide association study.	Anti-saccade response	antisaccade response measurement	• 4469 European	-	5	NA

Showing 1 to 5 of 351 rows5 rows per page

12345...71

Traits201

Search

Trait label(s)	Reported trait(s)	Association count with 2q37.1
bilirubin measurement	Bilirubin levels, Total bilirubin levels, Liver disease biomarker	36

GWAS Catalog

Trait → Tables of associations and studies

Trait label ⓘ

EFO ID ⓘ

Synonyms

Mapped terms ⓘ

Description

Reported Traits ⓘ

Child traits ⓘ

Highlighted Study ⓘ

breast carcinoma

EFO_000305

17 synonyms +

-

A carcinoma arising from the breast, most commonly the terminal ductal-lobular unit. It is the most common malignant tumor in females. Risk factors include country of birth, family history, menstrual and reproductive history, fibrocystic disease and epithelial hyperplasia, exogenous estrogens, contraceptive agents, and ionizing radiation. The vast majority of breast carcinomas are adenocarcinomas (ductal or lobular). Breast carcinoma spreads by direct invasion, by the lymphatic route, and by the blood vessel route. The most common site of lymph node involvement is the axilla. +

50 reported traits +

11 child traits +

Trait in OLS ↗

Trait in OXO ↗

Trait in Open Targets ↗

Trait in PGS Catalog ↗

Available data: Associations 1859 Studies 129 LocusZoom

Download Catalog data ↗

Include child trait data

Associations 1859

Search

↗

👤

Variant and risk allele	P-value	P-value annotation	RAF	OR	Beta	CI	Mapped gene	Reported trait	Trait(s)	Study accession	Location
rs4861477-C	8 x 10 ⁻⁶		0.15	-	-	-	AC093840.1, LINC02500	Response to tamoxifen in oestrogen receptor positive/HER2 negative breast cancer	response to tamoxifen, HER2 status, estrogen-receptor positive breast cancer	GCST006615	4:181490427
rs17198973-C	6 x 10 ⁻⁶		0.35	-	-	-	LINC02500, AC093840.1	Response to tamoxifen in oestrogen receptor positive/HER2 negative breast cancer	response to tamoxifen, HER2 status, estrogen-receptor positive breast cancer	GCST006615	4:181486725
rs10490113-?	5 x 10 ⁻⁶		NR	-	-	-	AC007179.2, AC007100.1, AC007092.1, LINC01783	Breast cancer	breast carcinoma	GCST000079	2:59272212
rs1876206-?	6 x 10 ⁻⁶		NR	-	-	-	FBN1	Breast cancer	breast carcinoma	GCST000079	15:48608389
rs1154865-?	7 x 10 ⁻⁷		NR	-	-	-	AC087879.1, LINC02444	Breast cancer	breast carcinoma	GCST000079	12:73596057

Showing 1 to 5 of 1859 rows 5 rows per page

Studies 129

Search

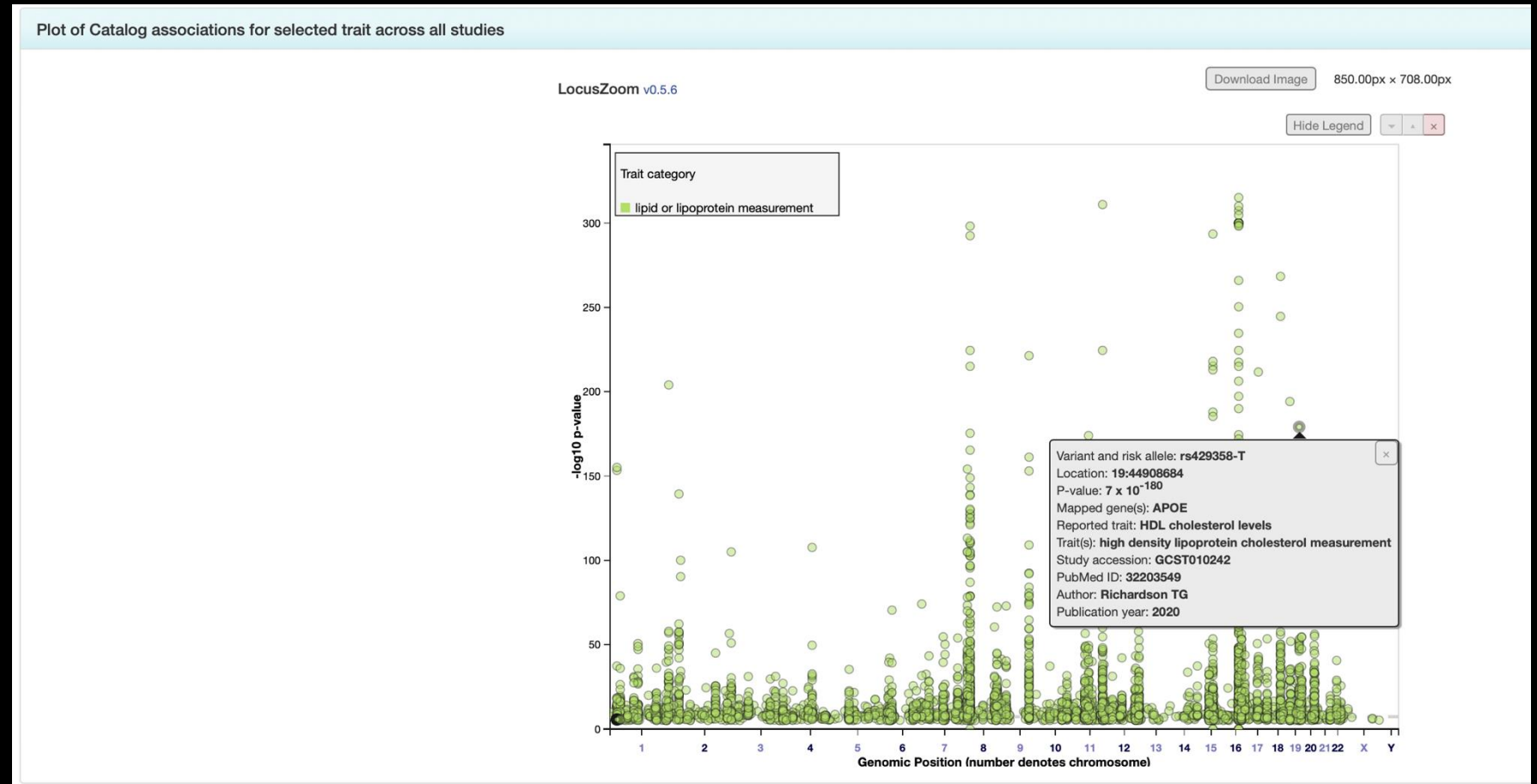
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👤

First author	Study accession	Publication date ⓘ	Journal	Title	Reported trait	Trait(s)	Discovery sample number and ancestry	Replication sample number and ancestry	Association count	Summary statistics
Fagerholm R	GCST004228	2017-03-14	Oncotarget	TP53-based interaction analysis identifies cis-eQTL variants for TP53BP2, FBXO28, and FAM53A that associate with survival and treatment outcome in breast cancer.	Survival in TP53 positive breast cancer	overall survival, TP53 Positive Breast Carcinoma	• 157 European	-	0	NA
Onishi H	GCST006615	2018-08-30	PLoS One	A genome-wide association study identifies three novel genetic markers for response to tamoxifen: A prospective multicenter study.	Response to tamoxifen in oestrogen receptor positive/HER2 negative	response to tamoxifen, HER2 status, estrogen-	• 275 East Asian	• 72 East Asian, Asian unspecified	2	NA

GWAS Catalog

Trait → interactive cross-study Manhattan plot



GWAS Catalog

Study → tables of associations, related studies

GWAS

Publication

27494321

Publication information

PubMed ID

27494321

Journal

PLoS Genet

First author

Jones SE

Publication date

2016-08-05

View in PubMed

View in Europe PMC

Title

Genome-Wide Association Analyses in 128,266 Individuals Identifies New Morningness and Sleep Duration Loci.

Authors

Jones SE, Tyrrell J, Wood AR, Beaumont RN, Ruth KS... [Show more](#)

Full Summary Statistics

Available data:

Associations 34

Studies 6

Download Catalog data

Associations 34

Variant and risk allele	P-value	P-value annotation	RAF	OR	Beta	CI	Mapped gene	Reported trait	Trait(s)	Study accession	Location
rs516134-C	7 x 10 ⁻²⁸		0.03	-	0.081 unit increase	[0.059-0.103]	RNASEL	Chronotype	circadian rhythm	GCST003837	1:182584558
rs11162296-G	2 x 10 ⁻²¹		0.84	-	0.037 unit decrease	[0.027-0.047]	AC093433.2, PIGK	Chronotype	circadian rhythm	GCST003837	1:77234511
rs10157197-G	6 x 10 ⁻¹⁷		0.6	-	0.025 unit increase	[0.017-0.033]	C1orf54	Chronotype	circadian rhythm	GCST003837	1:150278225
rs372229746-G	8 x 10 ⁻¹⁵		0.55	-	0.028 unit increase	[0.018-0.038]	POLR2J3, RASA4B	Chronotype	circadian rhythm	GCST003837	7:102518368
rs75804782-T	1 x 10 ⁻¹⁴		0.88	-	0.03 unit increase	[0.018-0.042]	TRAF3IP1, RNU6-234P	Chronotype	circadian rhythm	GCST003837	2:238407402

Showing 1 to 5 of 34 rows 5 rows per page

Studies 6

Study accession	Reported trait	Trait(s)	Discovery sample number and ancestry	Replication sample number and ancestry	Association count	Summary statistics
GCST003837	Chronotype	circadian rhythm	• 127898 European	• 89283 NR	14	FTP Download or API access
GCST003839	Sleep duration	sleep duration	• 127573 European	• 47180 NR	3	FTP Download or API access
GCST003838	Morning vs. evening chronotype	circadian rhythm	• 114765 European	-	15	NA
GCST003840	Sleep duration (oversleepers vs undersleepers)	sleep duration	• 39082 European	-	2	NA
GCST006685	Sleep duration (oversleepers)	sleep duration	• 91306 European	-	0	FTP Download or API access

Showing 1 to 5 of 6 rows 5 rows per page

Lab

Either individually or in small groups, use the list of 10 lead SNPs from a GWAS and the supplied tutorial/exercise code (on Canvas).

- Tutorial
 - Use LDlinkR to find LD proxies
 - Write out the list of rsIDs
 - Upload to VEP and annotate them
 - Use HaploR to query HaploReg and RegulomeDB for the 10 SNPs, using various parameters
 - Play with LDlinkR a little more
- Exercise: trying to replicate/verify some claims from a COVID-19 GWAS